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Correspondence:
cohler@post.harvard.edu

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Research Article

Dominance of Natural Atmospheric CO_2 Dynamics: Falsification of Anthropogenic Attribution Through Mass-Balance Analysis, Isotopic Diagnostics, and Thermodynamic First Principles

Jonathan Cohler [1,] and Willie Soon [2]*

[1] *Cohler & Associates, Inc., Lexington, MA 02420, USA; cohler@post.harvard.edu*

[2] *Institute of Earth Physics and Space Science, 9400 Sopron, Hungary; soon.willie@epss.hu*

* *Correspondence: cohler@post.harvard.edu*

Abstract

Observed changes in atmospheric CO_2 concentration during the instrumental period, and isotopic signatures back to the Little Ice Age, are likely explained by natural biogeochemical feedbacks, with no detectable anthropogenic contribution. This is supported by mass-balance reservoir routing showing short CO_2 residence times of 3.5–4 years, stable Keeling plot intercepts of −13.2‰ over centuries demonstrating overwhelming biosphere dominance and undetectable fossil-fuel influence, unidirectional precedence of local temperatures → CO_2 across all timescales from the present to ~500 million years ago, and thermodynamic considerations that establish the physical meaninglessness of a global mean surface temperature (GMST). IPCC Bern Impulse Response Function (IRF) model parameters show instability across assessments, violate mass conservation, and introduce unsupported source discrimination. The global bomb-^{14}C pulse from nuclear testing exhibits clean single-exponential decay with e-folding time $\tau = 18.389$ yr ($R^2 = 99.825\%$) fitted to the full 1965–2019 global Hua et al. (2022) record, extending and tightening the $\tau \approx 17.2$ yr upper bound Koutsoyiannis (2024) obtained from the 1965–1985 subset; the two fits are self-consistent across their shared range, with the longer window exploiting the complete post-ban decay. This single functional form over 54 years directly falsifies the Bern model's multi-exponential structure, IPCC permanent airborne fraction, and multi-millennial CO_2 persistence claims. Critical examination of conflicting airborne fraction definitions reveals foundational inconsistencies in IPCC CO_2 attribution. These convergent lines establish that natural processes explain observed CO_2 variations with negligible human influence. A unified first-principles synthesis shows that a modest natural top-of-atmosphere radiative imbalance of ~2.6 W m^{-2} accounts for the observed rise from 2000 to 2025 in atmospheric CO_2 via temperature-driven oceanic outgassing and enhanced soil respiration, with no anthropogenic contribution detectable in the Keeling net-input signature over the instrumental or sclerosponge records.

Keywords: CO_2 residence time; $\delta^{13}C$ isotopic signature; Keeling plot; Bern model; Revelle factor; Biosphere dominance; Anthropogenic fingerprint; Global mean surface temperature; Thermodynamic intensive property; Natural climate variability

1. Introduction: Observed CO_2 and Temperature Patterns: Challenges to Anthropogenic Attribution

In ongoing scientific and policy discussions, the Earth's climate system is characterized by complex interactions among atmospheric, oceanic, and biospheric components, with atmospheric carbon dioxide (CO_2) concentration and temperature serving as central variables. Long-term observational records, including direct measurements of CO_2 from sites such as Mauna Loa and the South Pole, isotopic ratios ($\delta^{13}C$) from the Scripps Institution of Oceanography, and temperature proxies spanning geological epochs, reveal patterns that challenge prevailing attributions of recent changes primarily to anthropogenic emissions. These records exhibit rising CO_2 levels accompanied by declining $\delta^{13}C$ values and varying local temperature trends, yet detailed analyses

using straightforward physical models consistently indicate that natural processes, driven by thermodynamic mechanisms including solubility, respiration, biogeochemical fluxes, oceanic outgassing, and solar influences, provide a sufficient explanation for the observed variations with no detectable human role [22,77,78]. This natural explanation is one possible account that is fully consistent with all current high-precision instrumental measurements and thermodynamic first principles; paleoclimate proxies extending to the Little Ice Age remain supportive but are not required for the quantitative attribution of the post-1958 rise.

This synthesis employs transparent first-principles methodologies, including mass-balance reservoir routing applied to Mauna Loa and Barrow data [77] and Keeling plot analysis of $\delta^{13}C$ measurements and proxy reconstructions extending back to the Little Ice Age [78]. These are contrasted with the Intergovernmental Panel on Climate Change's (IPCC) Bern models and culminate in a unified first-principles synthesis showing that a modest natural top-of-atmosphere radiative imbalance of ~2.6 W m^{-2} quantitatively accounts for the observed CO_2 rise within the Loeb $\pm1\sigma$ CERES envelope under standard van 't Hoff and Arrhenius kinetics.

High-resolution local proxy records and site-specific modern observations from every major archive examined demonstrate unidirectional local temperature → CO_2 precedence on all timescales. Ice cores from Antarctica (Vostok, EPICA Dome C, WAIS Divide, Taylor Dome) and Greenland (GISP2) show local temperature consistently leading CO_2 by 200–1,000+ years across every glacial termination and millennial-scale event, with no exceptions [42,92,100]. A comprehensive review of the literature over the past 650,000–800,000 years confirms that atmospheric CO_2 variations generally follow changes in temperature and other climatic variables rather than preceding them, with no quantitative validation for CO_2 as the prime mover in glacial-interglacial cycles [92,115]. Extending this record over the full Phanerozoic eon (last 500 million years), multiple independent reconstructions pair $\delta^{18}O$-based local temperature proxies with CO_2 proxies (boron isotopes in foraminifera, stomatal density in fossil leaves, alkenone $\delta^{13}C$), showing temperature changes preceding CO_2 changes with characteristic lags on the order of millions of years [79]. The authors are unaware of any proxy record or observational period in which atmospheric CO_2 leads temperature. Modern observations reinforce the pattern: major volcanic eruptions drive rapid cooling followed by CO_2-growth slowdowns [41]; El Niño warming precedes CO_2 spikes by ~6 months [28,84,103]; OCO-2 satellite (NASA Orbiting Carbon Observatory 2) [35] and FLUXNET [5,6] tower data show regional CO_2 fluxes tracking local temperature gradients; and recovery sequences after volcanic, ENSO, and warming events all show temperature recovery preceding CO_2 recovery. These convergent lines of direct physical evidence support unidirectional causation from local temperatures to CO_2 on all timescales examined.

These methods are supplemented by broader evidence, including correlations with high-variability solar reconstructions, the absence of detectable perturbations during emission reductions (e.g., COVID-19 lockdowns), and artifacts introduced by data homogenization processes that systematically alter historical trends [22,24,117].

In addition to these carbon cycle and causality insights, the paper incorporates fundamental thermodynamic critiques that question the validity of GMST as a physical metric. By examining the intensive nature of temperature and the arbitrariness of averaging schemes in non-equilibrium systems, these analyses underscore the limitations of metrics central to many climate assessments [20,37]. The purpose of this work is to present a cohesive, evidence-based reassessment grounded in verifiable calculations and direct observational fits, highlighting how simple, solvable models rooted in established physics align closely with data, while more elaborate frameworks often rely on assumptions that lack empirical support. This integrated perspective demonstrates that natural biogeochemical feedbacks and variability provide complete, data-consistent explanations for observed CO_2 variations with no detectable anthropogenic contribution. This paper's scope is twofold: (i) the attribution of the observed atmospheric CO_2 rise, and (ii) the thermodynamic validity of the GMST metric that underpins IPCC warming, sensitivity, projection, attribution, and sea-level claims. It does not address the radiative-transfer properties of CO_2 itself, which are governed by independent spectroscopic physics and are logically separable from carbon-cycle attribution.

2. Centennial-to-Millennial-Scale Temperature Precedence: The Medieval Warm Period and Little Ice Age as Natural Climatic Anomalies

On centennial-to-millennial timescales, Soon et al. (2003) [116] conducted an exhaustive reappraisal of more than 100 independent local and regional climate proxies spanning both hemispheres, establishing the Medieval Warm Period (≈800–1300 A.D.) and Little Ice Age (≈1300–1900 A.D.) as real, widespread climatic anomalies with global imprints. The study uses a locality-based paradigm that respects the distinct physical sensitivities, geographical constraints, and non-stationary calibration relations of each archive (ice cores, tree rings, boreholes, glaciers, corals, lake sediments, speleothems, historical documents, pollen, and phenological records), and demonstrates that these anomalies are not artifacts of physically meaningless global averaging but emerge consistently at the scale of individual sites and regions.

The Medieval Warm Period (MWP) is recorded as sustained warmth and/or reduced glacier extent at dozens of locations across Europe, North America, the North Atlantic, Asia, and parts of the Southern Hemisphere, enabling Viking settlement in Greenland, northward expansion of Mediterranean flora into central Europe, and favorable agricultural conditions. The subsequent Little Ice Age (LIA) is marked by widespread cooling, glacier advances, increased climatic variability, and societal impacts (e.g., abandonment of Norse settlements in Greenland, frequent severe winters in Europe). At the great majority of globally dispersed sites examined, intervals during the MWP were as warm as, or warmer than, any 50-year period within the 20th century, despite atmospheric CO_2 remaining stable near the pre-industrial value of ≈280 ppm.

Crucially, the 20th-century warming does not emerge as the warmest or most extreme period of the last 1000 years at most locations. Where 20th-century warmth is prominent, it often begins early (1920s–1950s) when cumulative anthropogenic CO_2 was still small. The study explicitly cautions against forcing disparate local proxies into simple hemispheric or global quantitative composites, noting that thermal and dynamical constraints imposed by local geography become increasingly dominant on 50–100+ year timescales. This "locality paradigm" directly reinforces the unidirectional local temperature → CO_2 precedence documented on shorter (months to millennia) and longer (Phanerozoic) timescales: major natural temperature excursions clearly occurred first, with no evidence that changes in atmospheric CO_2 initiated or drove them. These centennial-scale oscillations are fully consistent with natural solar and ocean-circulation variability.

Criticisms of Soon et al. (2003) [116] were predicated on the erroneous assumption that only synoptic global or hemispheric temperature composites constitute valid evidence of climatic anomalies—an assumption contradicted by the physical reality that temperature is an intensive property unsuitable for such averaging across Earth's heterogeneous non-equilibrium system. By deliberately avoiding both global averaging schemes and uncertain proxy-to-temperature conversions, Soon et al. (2003) adopted the scientifically supportable methodology. In contrast, multi-proxy global reconstructions [94,95] imposed such invalid averaging and have also been shown to suffer from significant statistical artifacts [96]. Subsequent thermodynamic analyses by Essex et al. (2007) [37], together with extensions by Cohler (2025) [20] and the framework developed in the present work, confirm that this locality paradigm is the appropriate approach, while the global composite methods underlying the Mann et al. (1998, 1999) reconstructions [94,95] are physically invalid and introduce non-physical artifacts. Criticisms of Soon et al. (2003) were based largely on the temperature averaging fallacies already mentioned rather than substantive engagement with its locality paradigm. Ironically, Soon et al. (2003) pioneered the systematic, site-specific spatial mapping of centennial-scale climatic anomalies using this locality-based approach—the very methodology that has since become the universally adopted standard in climate science, as seen in every modern NASA GISTEMP, HadCRUT, and IPCC regional anomaly 'heat' map.

3. Refined Reservoir Routing and Airborne Fraction Revisited: Logical vs. Computational Definitions and Biospheric Dominance

Analyses of the atmospheric carbon cycle using simple mass-balance reservoir models provide robust empirical evidence for short CO_2 residence times and the overwhelming dominance of natural processes [112]. In Koutsoyiannis' Refined Reservoir Routing (RRR) framework, the

atmosphere is treated as a dynamic reservoir governed by the continuity equation, where inflow and outflow are related through a power-law storage-outflow relationship, $Q = Q_0 (S/S_0)^b$. Optimization across extensive instrumental records from Mauna Loa (1958–present) and Barrow, Alaska (1961–present), yields an optimal exponent b=1 for outflow, corresponding to a strictly linear reservoir. This linearity produces excellent fits to observed CO_2 concentration time series, achieving explained variances exceeding 99.7% for storage (concentration) and approximately 85% for net inflow, with split-sample validation (fitting 1958–2002 and predicting 2003–2023) showing only minor drops in performance of 1–3.5% [77]. The resulting mean annual residence time is approximately 3.5–4 years (geometric mean accounting for seasonal variations ranging from 1.5 years in high-amplitude northern sites to 10 years), while the so-called "adjustment time," the characteristic timescale for system adjustment to perturbations, is equal to or shorter than the residence time in this linear configuration [50,51].

These short timescales arise directly from the rapid cycling of carbon through the biosphere and surface oceans, where annual natural fluxes (gross inflows and outflows) vastly exceed anthropogenic inputs. Natural processes account for approximately 96% of the total annual carbon turnover, while anthropogenic emissions represent only ~4% of the overall exchange. As a result, human-emitted CO_2 is quickly absorbed by natural sinks, with only ~6% of cumulative anthropogenic emissions remaining airborne [76,77].

This low value is explained by Koutsoyiannis (2024) [76,77], which notes and clarifies the common incorrect use and conflation of several different conflicting definitions of "airborne fraction." Koutsoyiannis defines his "logical airborne fraction" (ABF_L) the same way that IPCC AR6 defines "Airborne fraction":

> "*The fraction of total CO_2 emissions (from fossil fuels and land-use change) remaining in the atmosphere*" [60] (p. 2237)

In principle, this is a physically consistent, mass-conservation-based quantity tracking what portion of emitted anthropogenic mass accumulates after sink removal:

$$\mathrm{ABF_L} \equiv \frac{(\textbf{remaining}\ \text{anthropogenic atmospheric}\ \mathrm{CO_2})}{(\textbf{cumulative total}\ \text{anthropogenic emissions})} \tag{1}$$

Keeling (1973) [72] originally described this concept (attributing it to earlier ideas like Callendar, 1938, 1940 [16,17]) as the fraction of industrial CO_2 input that remains airborne, aligning with $\mathrm{ABF_L}$.

However, a second "computational airborne fraction" ($\mathrm{ABF_C}$), often misattributed to Keeling via Bolin (1977) [12] and used in IPCC assessments, although it contradicts the AR6 glossary entry (which is equivalent to $\mathrm{ABF_L}$), is calculated as:

$$\mathrm{ABF_C} \equiv \frac{(\textbf{annual}\ \text{increase in atmospheric}\ \mathrm{CO_2})}{(\textbf{annual}\ \text{anthropogenic emissions})} \tag{2}$$

which implicitly assumes the entire observed $\Delta[CO_2]$ is anthropogenic—circularly presupposing dominance and yielding historical estimates of ~40–100% (e.g., Callendar, 1938: 75%; Callendar, 1940: 100%; Ekdahl and Keeling, 1973: 49%; Bolin, 1977: 40 ± 5%; AR6: ~44%) [12,16,17,34,61]. This definition is different from and incompatible with $\mathrm{ABF_L}$, as it ignores natural variability and net fluxes, leading to overestimation of anthropogenic persistence.

In contrast, $\mathrm{ABF_L}$ is computed via the RRR framework, applying a mean residence time of $\tau =$ 4 years [77] to cumulative emissions. The remaining mass $\mathrm{M_R}$ (anthropogenic CO_2 remaining in the atmosphere) is given by the convolution integral assuming exponential decay:

$$\mathrm{M_R} = \int_{t_0}^{t_c} e^{(t-t_c)/\tau} dm_A(t) \tag{3}$$

where $dm_A(t)$ is the anthropogenic emission rate at time t, t_0 is the start year, t_c is the current year, and the exponential kernel reflects removal processes. Then for $t_0 = 1850$ and $t_c = 2023$, Koutsoyiannis [76,77] calculated:

$$\mathrm{ABF_L} = \frac{\mathrm{M_R}}{\mathrm{M_A}} = \frac{\int_{t_0}^{t_c} e^{(t-t_c)/\tau} dm_A(t)}{\int_{t_0}^{t_c} dm_A(t)} \approx \frac{163\ \mathrm{GtCO_2}}{2612\ \mathrm{GtCO_2}} = \frac{20.9\ \mathrm{ppm}}{334.9\ \mathrm{ppm}} \approx 6\% \quad (4)$$

This indicates ~94% of cumulative anthropogenic emissions (expressed in $GtCO_2$ or GtC)[1] have been absorbed by natural sinks on short timescales. Stallinga (2023) [118] independently estimates $\mathrm{ABF_L} \approx 12\%$, which is also significantly lower than the AR6's $\mathrm{ABF_C}$ estimate of 44%.

Under this same RRR framework of Koutsoyiannis, a single hypothetical ~334.9 ppm CO_2 pulse would decay below 1 ppm within approximately 23 years if emissions stop, implying no detectable persistence over centuries or millennia [22,50,51,77]. Independent physics-based modeling similarly derives residence times of ~4 years via local thermodynamic amplification (temperature-dependent solubility per Henry's law and enhanced respiration/outgassing) dominating the cycle [50,51,118] (see also Table 1).

4. The Bern Model and the Residence vs. Adjustment Time Distinction

We first contrast empirical residence-time evidence with the IPCC's adjustment-time construct, demonstrate first-principles violations, then examine the Revelle misapplication, Surface Ocean CO_2 Atlas (SOCAT) data gaps, stylized carbon-budget schematics, and the resulting millennial-persistence artifact.

The Bern model and similar multi-compartment representations underpinning IPCC assessments posit prolonged adjustment times (centuries to millennia) with 14%–22% of CO_2 input to the atmosphere persisting indefinitely via a non-decaying a_0 term, derived from theoretical parallel reservoirs, slow exchange rates, and nonlinear carbonate buffering [67,68]. This a_0 term is fundamentally non-physical: it excludes ultimate geological sinks such as silicate weathering and sediment burial, leading to indefinite retention of the a_0 fraction of every emission and unbounded accumulation under continued emissions and contradicting Earth's observed long-term carbon cycle [77,112]. The Bern IRFs' coefficients and timescales are not calibrated to any carbon-cycle observation; they are fitted to the pulse-response output of CMIP ESMs that are themselves tuned primarily to reproduce historical GMST. The thermodynamic invalidity of that tuning target (see Section 5.2, below) is therefore inherited by every IRF-derived quantity.

The Bern model falsely assumes the ocean consists of *parallel* compartments directly connected to the atmosphere, employing 3 to 5 exponential terms plus one permanent (non-decaying) term in AR2–AR6 (see detail below). In reality, CO_2 can enter the ocean only at the surface; the instant molecules dissolve there, they are already removed from the atmosphere. All slower timescales (surface → thermocline → deep ocean) are serial and internal to the ocean. We refer to this failure of the Bern model as the "parallel-serial fallacy." There is no global-scale empirical evidence of CO_2 saturation at any ocean depth despite the purely model-based claim of Jiang et al. (2019, 2023) [64,65], whose reported "global" increase in the Revelle Factor is derived from statistical interpolation and Earth System Models (ESM) applied to sparse ship-track data. The Revelle Factor is a strictly local intensive property, like temperature or pH, and therefore cannot be meaningfully averaged across the ocean's heterogeneous, non-equilibrium water masses and vastly different spatial and temporal scales.

Although saturation would in theory be gradual, evidence shows that current absorption rates are not slowing. Relative land and ocean absorption rates have stayed proportional to atmospheric CO_2 for 270 years despite a 50% rise in pCO_2 Müller (2025) [98]. These empirical indicators render the Bern models' "long tails" unverifiable assumptions rather than data-supported features. Linear reservoir approaches, with their simplicity and high fidelity, show that natural biogeochemical processes quantitatively account for observed CO_2 dynamics within the Loeb ±1σ CERES envelope, with human contributions negligible, transient, and rapidly integrated into the dominant bulk natural cycle.

1. In this paper, we use both GtC and $GtCO_2$ to describe the amount of carbon in all the so-called carbon reservoirs. To convert from GtC to $GtCO_2$ multiply by 44/12 ≅ 3.67.

4.1 Residence Time, Adjustment Time, and the Bern Impulse-Response Function

The IPCC defines "lifetime" (or "residence time") as the empirically measurable quantity given by "the total mass of the species in the reservoir divided by the total rate of removal from the reservoir," a direct ratio grounded in observable atmospheric burden and flux data [60]. The report also defines "turnover time" as "[t]he time it takes to replenish the content of a reservoir," calculated as "the ratio of the content of a reservoir to the rate of input or output (inflow or outflow) from that reservoir," and explicitly notes that turnover time "is equivalent to the lifetime when the reservoir is in steady state" [60]. These terms thus correspond to the physically meaningful and empirically verifiable residence time (~3.5–5 years) commonly used in the literature.

In contrast, "response time or adjustment time" is defined purely theoretically as "[t]he time scale characterizing the decay of an instantaneous pulse input into the reservoir." It is further specified as the time taken for the concentration to decline to 1/e (~37%; e-folding time) of its original value following such a hypothetical perturbation. This is a construct reliant on unobservable, model-dependent "instantaneous pulse inputs" with no direct empirical manifestation in the real world other than the "natural" experiment provided by bomb-radiocarbon decay [23,54]. This distinction exists only within the Bern model's multi-compartment assumptions and disappears in data-driven linear frameworks (see, e.g., Cohler & Soon [23]).

4.1.1 Falsification by Transparent Linear Models

The concept of a short atmospheric CO_2 residence (turnover) time has been consistently documented in the peer-reviewed literature for decades. Sundquist (1985) [124] first compiled measured residence times from numerous independent studies, revealing values predominantly in the 5–15-year range. Segalstad (1998) [112] expanded this compilation into a comprehensive table of 25 distinct determinations (predominantly 5–10 years) and explicitly rejected any distinction between this directly observed turnover time and a purportedly much longer "adjustment time" or "response time," noting that longer theoretical values derived from nonlinear models contradict the measurable physical reality and are therefore irrelevant. Despite this long-standing empirical consensus, the IPCC has continued to assert that while individual CO_2 molecules exhibit a turnover time of ~4 years, so-called 'excess' anthropogenic CO_2 possesses a separate and far longer adjustment time of centuries to millennia. Outside the primary Joos et al. impulse-response lineage, this distinction has been substantively defended in mainly three papers: Maier-Reimer & Hasselmann (1987) [93], Archer et al. (2009) [2], and Sonnemann (2013) [114].

Table 1 extends the work of Segalstad to the present day and demonstrates that transparent, data-driven linear frameworks and analytically solvable models not only confirm short, equivalent residence and adjustment times but, using first principles and high-fidelity observational fits, also directly falsify each specific mechanism invoked in those three papers to justify a separate long adjustment time (see Sections 3, 4.1 here, and 4.2 for detailed refutations).

Furthermore, these studies explicitly falsify the IPCC residence-adjustment distinction and the Bern framework. Using the Koutsoyiannis (2024) [77] refined reservoir routing (RRR) model optimized on Mauna Loa and Barrow data, the strictly linear configuration (exponent b=1) yields explained variances exceeding 99.7% for concentration (and ~85% for net inflow), with mean annual residence and adjustment times both ~3.5–4 years (geometric mean accounting for seasonal variation) and full exponential decay without long tails. Split-sample validation (1958–2002 fit predicting 2003–2023) shows negligible performance drop, confirming robustness absent unverifiable compartments.

Müller (2025) [98] uses mass-balance analysis and the observed constancy of relative land and ocean CO_2 absorption over 270 years, proportional to atmospheric CO_2 concentration via Henry's Law and biomass linearity, to show that the residence time of atmospheric CO_2 is ~4 years. Added CO_2 distributes proportionally among the reservoirs in equilibrium (communicating-tubes principle), rendering claimed residence times of 30 years or more for the anthropogenic component physically inconsistent with observed proportionality, the equivalence principle, and established laws.

Stallinga (2023) [118] demonstrates mathematically that in any first-order linear system with parallel sinks, consistent with rapid natural cycling, adjustment time cannot exceed residence time

and typically equals it, rendering the claimed centuries-long delay highly improbable without artificial cascaded constraints.

Berry (2019 and 2021) [10,11] replicate the natural carbon cycle via physics-based inflow-outflow balance and shows that human emissions equilibrate rapidly: the human atmospheric fraction matches the human fraction of total inflow, with adjustment identical to residence time and no mechanism for indefinite persistence. Bomb-^{14}C decay further exposes the fallacy: Bern multi-compartment models require complex tuning, yet simpler linear approaches fit the curve with short overall τ (~17 years for the tracer, still orders of magnitude below millennia claims) without long tails [77,112].

Koutsoyiannis (2024) [77] fit the 1965–1985 subset to obtain $\tau \approx 17.2$ yr as an upper bound; the $\tau = 18.389$ yr value used in Figure 1 and elsewhere in this paper extends that fit to the full 1965–2019 record, with the two fits remaining self-consistent across their shared range.

Essenhigh (2009) [36] derives from basic mass-balance equations that short residence times preclude significant anthropogenic accumulation, as so-called 'excess' CO_2 is removed on the same timescale as natural fluxes dominate (96% of turnover).

Harde (2017, 2019, 2023) [50–52] similarly calculate effective residence times of ~3–4 years from ^{14}C constraints and parallel uptake processes, with adjustment times collapsing to short values; Bern multi-box models artificially prolong times by assuming slow, sequential compartments unsupported by data fits.

In these frameworks with an exponential decay timescale (e-folding time) of $\tau \approx 4$ years, the expected anthropogenic CO_2 contribution is calculated as follows: cumulative human emissions since pre-industrial times (~1850) total approximately 2500 $GtCO_2$ (gigatonnes of CO_2, from fossil fuels and land-use changes, per standard Global Carbon Project and IPCC estimates); adding roughly 7.82 $GtCO_2$ raises atmospheric concentration by 1 ppm (using standard conversion factor based on the total mass of the atmosphere and CO_2 molar mass); thus, 2,500 $GtCO_2 \div 7.82$ $GtCO_2$ per ppm yields a hypothetical excess of ~320 ppm if natural sinks absorb none of the additions. These models predict that this hypothetical excess would decay below 1 ppm within ~23 years ($t \approx 4 \times \ln(320) \approx 23$ years), implying no detectable persistence and directly contradicting Bern projections.

Simpler models reproduce key observables (linearly increasing absolute uptake, absence of saturation or slowdown) without nonlinear terms or unverifiable assumptions. Scenario tests underscore this: excluding anthropogenic inputs captures dynamics via temperature-driven natural variation, while anthropogenic-only terms fail entirely [51,77]. Conversely, Bern-type models predict observable uptake slowdowns and accelerating accumulation not seen in the observational record.

The long adjustment times and persistent fractions are thus model artifacts, not empirical features, repeatedly falsified by superior data alignment of linear approaches [10, 11, 23, 35, 49, 50, 76, 97, 111, 117]. The common argument, frequently advanced in defense of IPCC positions, that short residence time estimates reflect a confusion with a distinct "adjustment time" is circular: the distinction relies on the assumptions of Bern multi-compartment models that are contradicted by simpler, data-consistent frameworks, thereby prioritizing theoretical constructs over direct observational fidelity.

4.1.2 First-Principles Violations: Cascaded Sinks and Molecular Indistinguishability

The Bern model's structure fundamentally violates these first-principles realities. In IPCC AR2, AR3, AR4, AR5, and some sections of AR6 [55–57,59,61], the effective impulse response function for an idealized CO_2 pulse was given as

$$f(t) = a_0 + \sum_{i=1}^{n} a_i \ \mathrm{e}^{-t/\tau_i} \tag{5}$$

where a_0 was parameterized as 0.1369 in AR2 (five exponential modes), 0.152 in AR3, 0.217 in AR4, and 0.2173 in AR5/AR6 (three exponential modes each; see corrected values below) with

non-zero a_0 representing a non-decaying fraction persisting indefinitely (effective $\tau_0 = \infty$ and the summed exponential terms capturing multi-timescale decay with τ_i ranging from one to hundreds of years [1,62]. This cascaded form implicitly assumes parallel, source-discriminating sinks where so-called 'excess' (anthropogenic) CO_2 is somehow absorbed separately, with portions irreversibly retained in the atmosphere or slowly re-equilibrated due to assumed saturation and compartmental inertia. Because CO_2 molecules are chemically identical regardless of origin, sinks cannot selectively retain "anthropogenic" CO_2. All removal occurs on the single atmospheric reservoir at a unified short timescale $\tau_{total} \approx 3.5–5$ years (see Table 1) dominated by natural fluxes [77,118].

$$\frac{1}{\tau_{total}} = \sum \frac{1}{\tau_i} \tag{6}$$

A further and equally important inconsistency arises from the strictly serial nature of real ocean carbon transport, i.e. the Bern model's "parallel-serial fallacy." Any CO_2 molecule that reaches the deep ocean must physically pass through the surface mixed layer, then the thermocline, then intermediate depths in sequence. No physical pathway exists for a fraction of molecules to travel directly from the atmosphere to the deep ocean on centennial-to-millennial timescales.

Yet AR5 [59] Box 6.1 presents the Bern impulse-response function (IRF) as a description of physical processes, stating that "about 15 to 40% of CO_2 emitted until 2100 will remain in the atmosphere longer than 1000 years" and that "this second phase will pull the remaining atmospheric CO_2 fraction down to 10 to 25% of the original CO_2 pulse after about [10,000 years]" (AR5 Box 6.1; [1,68]). The entire Box 6.1, including the three-phase diagram, the table of removal processes, and the explicit attribution of each timescale to land uptake, ocean invasion, carbonate reaction, and silicate weathering, is simply the Bern model translated into words and figures. Even Joos et al. (2013) [68] explicitly cautioned that the coefficients a_i and timescales τ_i "have no direct process-based meaning, but are fitting parameters." Nevertheless, AR6 [61] attributes mechanistic meaning to the terms and relies on the resulting long adjustment-time tails to justify its claims of multi-millennial CO_2 persistence, remaining carbon budgets, and Zero Emissions Commitment [62] (Chapter 5). Compounding the non-physical scenario, these very parameters are not derived from observed ocean physics or first principles; they are obtained by fitting to the outputs of complex ESMs that are themselves tuned primarily to reproduce the thermodynamically invalid GMST metric over short historical periods. The Bern framework is therefore a self-referential circular construction: invalid temperature averaging → GMST-calibrated GCMs → phenomenological short-time-scale curve-fit → long-term persistence claims presented to policymakers as established physical fact. This parallel-serial fallacy renders the model's primary long-term claim, the multi-century to millennial adjustment-time tails and permanent a_0 fraction that drive future CO_2 emission scenarios, structurally impossible on first-principles grounds.

The instrumental record, which exhibits proportional uptake, and no multi-timescale residuals, confirms a single, short effective rate without need for cascaded discrimination or indefinite persistence. The Bern model's assumptions are thus not only non-physical but empirically falsified, underscoring that simple frameworks provide the only consistent explanation.

4.1.3 Parameter Instability of the Bern Impulse Response Function Between IPCC Reports

The evolution of the Bern model's impulse response function across IPCC reports illustrates persistent use of unverifiable, non-physical assumptions despite mounting contradictions from simpler, data-consistent approaches.

In IPCC AR2 [55], the Bern-SAR version was employed, approximating CO_2 decay with multi-exponential forms using a non-decaying a_0 term of 13.7% and featuring five different decaying terms with time scales up to 371.6 years [69].

$$f(t) = 0.1369 + 0.1298\, e^{-t/371.6} + 0.1938\, e^{-t/55.7} + 0.2502\, e^{-t/17.01} + 0.2086\, e^{-t/4.16} + 0.0807\, e^{-t/1.33} \tag{7}$$

Table 1. This table updates and extends the classic compilations of Sundquist (1985) and Segalstad (1998), which reported short residence times (predominantly 5–15 years) derived from multiple methods including bomb-^{14}C decay and explicitly rejected any distinction between this directly measured turnover time and a purportedly much longer "adjustment time" or "response time." All modern studies employing transparent linear or first-principles models contradict the IPCC position of a short molecular turnover time (~4 years) existing separately from a centuries-to-millennia adjustment time for 'excess' anthropogenic CO_2. The only three papers outside the Joos et al. impulse-response framework that substantively defended a separate long adjustment time (Maier-Reimer & Hasselmann 1987; Archer et al. 2009; Sonnemann 2013) are each directly falsified in this paper using Refined Reservoir Routing fits to Mauna Loa and Barrow data, ocean circulation realities, and first-principles arguments.

Source	Residence, Turnover, Lifetime in years	IPCC-like Long Adjustment Time in years	Falsified in this paper
Cohler & Soon (2026) [23]	3.8	—	—
Müller (2025) [98]	~4	—	—
Koutsoyiannis (2024) [77]	~3.5–4	—	—
Harde (2023) [52]	3.8	—	—
Stallinga (2023) [118]	~5	—	—
IPCC AR6 Annex VII Glossary (2021) [60]	~4	15–40%; >1000 10–25%; 1000s rest; $>10^4$ to $>10^5$	**Section 4.1:** (non-physical; pulse construct; serial-vs-parallel mismatch; molecular indistinguishability)
Berry (2021) [11]	3.5	—	—
Berry (2019) [10]	4–5	—	—
Harde (2017, 2019) [50,51]	4	—	—
Sonnemann & Grygalashvyly (2013) [114]	—	130 "equilibrium lifetime"	**Section 3:** (RRR linear model gives superior >99.7% variance fit at $\tau \approx 3.5$–4 yr with adjustment = residence); **Section 4.1:** (long-adjustment empirical constructs collapse in transparent linear frameworks)
Archer et al. (2009) [2]	—	20–35% 200–2000	**Section 4.1:** (pulse emission thought experiment non-physical; serial ocean transport impossibility); **Section 4.2:** (Revelle is local surface effect, not global barrier; rapid surface circulation feedbacks dominate)
Essenhigh (2009) [36]	~5	—	—
Maier-Reimer & Hasselmann (1987) [93]	—	100s–1000s	**Section 4.1:** (violates real serial ocean pathway requirement; multi-compartment construct impossible on first principles); **Section 4.2:** (ocean circulation dominated by fast surface currents, not slow serial compartments)

In AR3 and AR4 [56,57] the Bern2.5CC model was used, with the permanent fraction now parameterized as ~15% and ~22% successively, tuned to historical airborne fractions over ~150–200 years of data [58,70]. The longest time scale dropped 54% from ~370 years to ~170 years.

$$f(t) = 0.152 + 0.253\, e^{-t/171.0} + 0.279\, e^{-t/18.0} + 0.316\, e^{-t/2.57} \quad (8a)$$

$$f(t) = 0.217 + 0.259\, e^{-t/172.9} + 0.338\, e^{-t/18.51} + 0.186\, e^{-t/1.186} \quad (8b)$$

AR5 [59] updated to, and then AR6 inherited, a multi-model mean based on Joos et al. (2013) [68], keeping the permanent fraction at ~22%, but raising the longest time scale 128% from ~170 years to ~394 years.

$$f(t) = 0.2173 + 0.2240\, e^{-t/394.4} + 0.2824\, e^{-t/36.54} + 0.2763\, e^{-t/4.304} \quad (9)$$

The coefficients (a_i) and timescales (τ_i) were derived by fitting the analytical multi-exponential form (plus a constant non-decaying term) to the numerical pulse-response outputs of simplified 1D, 2D, intermediate complexity, and complex 3D ocean general circulation models and, in 2013, to multi-model ensemble means from ESMs of intermediate complexity (EMICs) [36,37,54,55]. The IPCC directly adopted these fitted Bern IRFs starting from AR2 [55] onward, propagating the parent models' unverifiable assumptions of compartmental bottlenecks, buffering limitations, and long-term persistence into official assessments of carbon cycle projections, airborne fraction estimates, and greenhouse gas metrics in IPCC AR3, AR4, and AR5/AR6 [56,57,59].

The Bern impulse-response function is not calibrated to any real-world observations of the carbon cycle. It is a purely mathematical fitting function calibrated exclusively to the simulated CO_2 decay curves produced by those models [68]. Moreover, these "model-generated output curves" are not even raw individual runs anymore: the final Bern IRF used by the IPCC is fitted to the multi-model ensemble mean—the simple arithmetic average of all participating models, each given equal weight despite their vastly different complexities and shared structural assumptions [68] (Section 3.2). Because the models are neither independent nor identically distributed and are tuned primarily to the physically meaningless GMST metric, this ensemble averaging adds yet another layer of invalid statistical abstraction.

Joos and colleagues have been explicit about the limitation: the individual timescales and coefficients "have no direct process-based meaning, but are fitting parameters" [68] and "should be interpreted primarily as abstract fitting parameters" [122]. Neither Joos nor the IPCC has ever justified quoting these tunable parameters to three and four significant digits, nor explained the chaotic, non-convergent jumps they exhibit across reports. A detailed examination of these four impulse-response functions (Table 2) reveals extreme structural instability in e-folding times, weighting coefficients, and timescales.

Bern impulse response function (IRF) coefficients. AR2 values are the Bern SAR parameterization (Joos et al. 1996 [69]); AR3/AR4 values are the Bern 2.5CC parameterization (Joos et al. 2001 [70]); AR5 values are from Joos et al. (2013) [68] Table 5, reproduced unchanged in AR6 WG1 Chapter 7 and used in Table 7.SM.7 to compute CO_2 AGWP at 20, 100, and 500-year horizons. The source paper states explicitly that the a_i and τ_i "have no direct process-based meaning, but are fitting parameters" [68], and limits the fit to $0 \leq t \leq 1000$ yr. No per-coefficient physical process assignment is made in any IPCC Assessment Report. AR5 WG1 Box 6.1 associates four generic process categories (photosynthesis-respiration, "seawater buffering," reaction with $CaCO_3$, silicate weathering) with coarse timescale ranges spanning 1 to 10^6 yr, but these are not mapped to individual Bern coefficients. The "seawater buffering" category rests on extrapolating the Revelle factor, a local intensive surface property measured at a sparse handful of surface stations and defined only for the ocean mixed layer, to a global rate limit on bulk ocean CO_2 uptake. No such global constraint has ever been demonstrated. The Earth's geological record of recovery from massive non-fossil-fuel CO_2 injection events (for example the Paleocene-Eocene Thermal Maximum, with roughly 3,000 to 7,000 GtC absorbed and atmospheric CO_2 returned to baseline within approximately 100 to 200 kyr) directly falsifies the premise. The non-decaying mode a_0 lies outside the fit's own 0-to-1000-yr validity range, yet it inflates every AGWP computed from this IRF, including at the 100- and 500-year horizons used in AR6. The "Instability" column in Table 2 reports the step-to-step change of each τ_i across successive reports. The magnitudes

shown (timescale swings up to +263%) are retunings of a non-physical fit, not evolving physical understanding.

Claims that a substantial fraction of fossil-fuel CO_2 will persist in the atmosphere for 10,000 to 100,000 years, routinely repeated in IPCC-era communications, trace to a very small set of reduced-complexity simulations (Archer et al. 2009 [2]). No comprehensive Earth System Model has ever been run for 10,000 years, let alone 100,000. The long-tail runs on which these claims rest use EMICs and box models (CLIMBER-2, UVic, GENIE, MESMO, GEOCYC) that Archer's own paper acknowledges do not reach steady state. The 100,000-year persistence figure is not a simulation output. It is an input *assumption*: Archer specifies that 10% of the CO_2 persists until silicate weathering removes it on a 100,000-year timescale. The long-tail mechanism rests on (i) the Revelle-factor extrapolation refuted above, (ii) parameterized $CaCO_3$ dissolution, and (iii) a parameterized silicate-weathering sink. None of these has empirical support. All are contradicted by the geological record of complete atmospheric CO_2 drawdown following every non-fossil-fuel high-CO_2 episode in Earth's history.

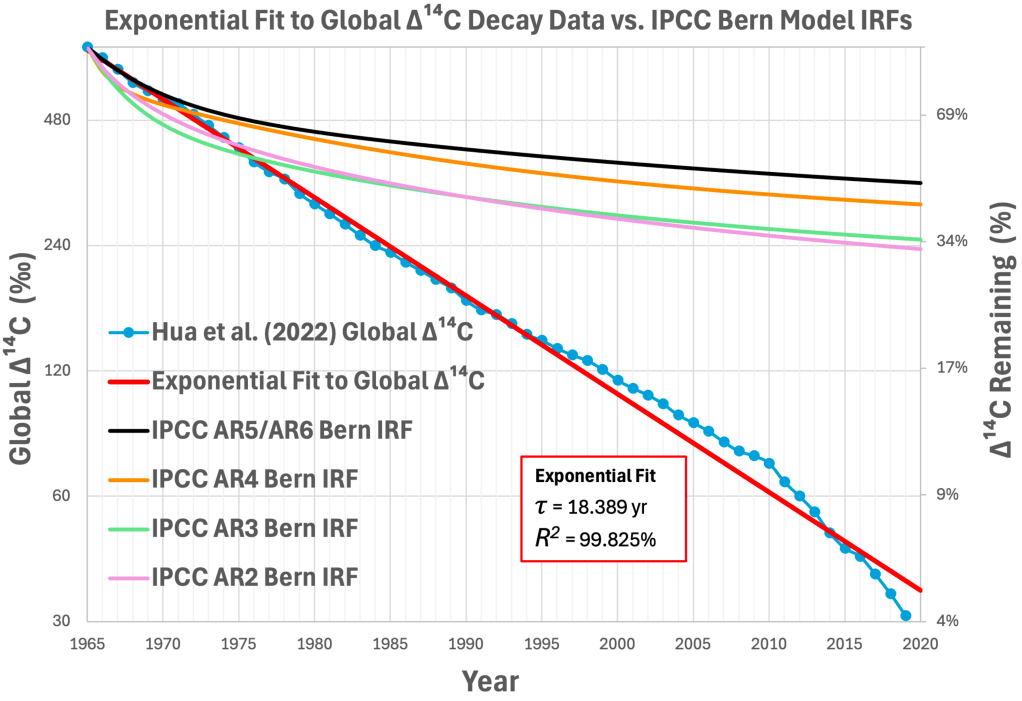


Figure 1. *Observed global Δ¹⁴C bomb-pulse decay from Hua et al. (2022) [54] (blue) compared with the AR2, AR3, AR4, and AR5/AR6 IPCC Bern impulse-response functions (purple, green, orange, black) and a single-exponential non-linear least-squares fit (red) with τ = 18.389 yr and R² = 99.825% over the full 1965–2019 global time series. On a semi-log scale, a single-exponential function displays as a straight line while any sum of positive-weighted exponentials displays as an upward curve. All IPCC Bern IRFs deviate significantly from observation throughout the decay record with ever-growing discrepancy after an initial crossover that occurs between years 2 and 11 depending on the parameterization. By 2019, the observed Δ¹⁴C is 4.3% of its 1965 peak value, while the AR2, AR3, AR4, and AR5/AR6 Bern IRFs predict 33%, 35%, 42%, and 48% respectively. The slightly higher τ obtained here (18.389 yr) relative to the 17.2 yr value mentioned in his paper reflects that Koutsoyiannis (2024) [77], Appendix C fits the 1965–1985 subset of the same series; the single-exponential form of the Hua et al. (2022) record is however preserved across the full 1965–2019 range, so the two fits are mutually consistent.*

Such ad hoc restructuring, including unexplained mode disappearances, timescale swings by factors of 2–4, and coefficient jumps of 40%–95%, is incompatible with physical realism, which is objective and invariant (see Table 2). If the Bern model genuinely captured discoverable properties of the carbon cycle, successive refinements would drive convergence toward stable values rather than this pattern of large-magnitude uncontrolled oscillations. The a_0 term arises solely from (i) deliberately omitting geological sinks (silicate weathering and sediment burial) and (ii) fitting to finite-length model runs (3 comprehensive ESMs run for only the first 100 years; 10 EMICs reaching approximately 1,000 years) that never reach full atmosphere, ocean, geosphere

equilibration. In the real Earth system, these ultimate sinks have repeatedly drawn atmospheric CO_2 down from levels of several thousand ppm to pre-industrial values without leaving any permanent airborne residue.

Table 2. Evolution and structural instability of the Bern impulse-response function parameters and e-folding times across IPCC Assessment Reports (AR2–AR6). Timescales τ_i are in years; weighting coefficients a_i are shown in parentheses; e-folding times are shown in square brackets. Instability percent changes are relative to the preceding report (AR2→AR3→AR4→AR5/AR6) with anomalous changes indicated in bold.

Mode	AR2	AR3	AR4	AR5/AR6	Instability
[e-folding]	[41.2 yrs]	[45.6 yrs]	[95.6 yrs]	[165 yrs]	+11%, **+110%**, **+73%**
Permanent (a_0)	∞ yrs (0.1369)	∞ yrs (0.152)	∞ yrs (0.217)	∞ yrs (0.2173)	+11%, **+43%**, +0.14%
Long (a_1)	371.6 yrs (0.1298)	171.0 yrs (0.253)	172.9 yrs (0.259)	394.4 yrs (0.2240)	**-54%**, +1.1%, **+128%** **+95%**, +2.3%, -14%
Intermediate (a_2)	55.7 yrs (0.1938)	18.0 yrs (0.279)	18.51 yrs (0.338)	36.54 yrs (0.2824)	**-68%**, +2.8%, **+97%** **+44%**, +21%, -16%
Short (a_3)	17.01 yrs (0.2502)	2.57 yrs (0.316)	1.186 yrs (0.186)	4.304 yrs (0.2763)	**-85%**, **-54%**, **+263%** +26%, **-41%**, **+49%**
Shorter (a_4)	4.16 yrs (0.2086)	— —	— —	— —	**AR2 only**
Shortest (a_5)	1.33 yrs (0.0807)	— —	— —	— —	**AR2 only**

This parameter instability reveals deeper deficiencies in the underlying ESMs and EMICs to which the Bern IRF is fitted. Those complex models, to which the Bern IRF is fitted, are tuned primarily to reproduce historical GMST trajectories (itself a physically meaningless quantity in Earth's heterogeneous non-equilibrium system [20,37]) using hundreds of adjustable, poorly constrained parameters. The ever-changing Bern coefficients and timescales are therefore the downstream symptom of a deeper circularity: a simple fitting function tuned to the output of other models that were themselves tuned to an invalid metric. When successive IPCC reports present these ad hoc, non-convergent parameters as quantitative representations of real carbon-cycle processes, the practice crosses from legitimate modeling into methodological circularity that undermines the scientific credibility of the entire long-term persistence narrative. The IPCC's Bern IRFs all contain a non-physical a_0 term that increased 59% from 0.1369 in AR2 [48] to 0.2173 in AR5/AR6 [68]). a_0 is a constant that enters the response uniformly at every time scale. The other exponential components contribute meaningfully only within a few multiples of their respective timescales. The constant a_0 has no such localization: it raises the atmospheric fraction at every horizon, starting at year 1 and including the 20-year, 100-year, and 500-year Global Warming Potentials used in UNFCCC national emissions inventories, and the 1000-year persistence claim illustrated in AR5 Box 6.1 Figure 1 [68]. a_0 is a fitting artifact produced by restricting the time domain of the underlying model runs to less than 1000 years. The Bern IRFs therefore do not represent CO_2 fate on geological time scales where silicate weathering and ocean carbonate sedimentation drive atmospheric CO_2 toward zero.

4.1.4 Bomb-^{14}C Decay as Empirical Falsification of the Bern Model

Atmospheric nuclear-weapons testing (1955–1963) created a sharp global pulse of excess $^{14}CO_2$. The global mean $\Delta^{14}C$ compilation of Hua et al. (2022) [54] reached its maximum of approximately 700 ‰ in 1965 [77] (see Figure 2). After atmospheric nuclear testing ceased, the excess declined under natural removal processes.

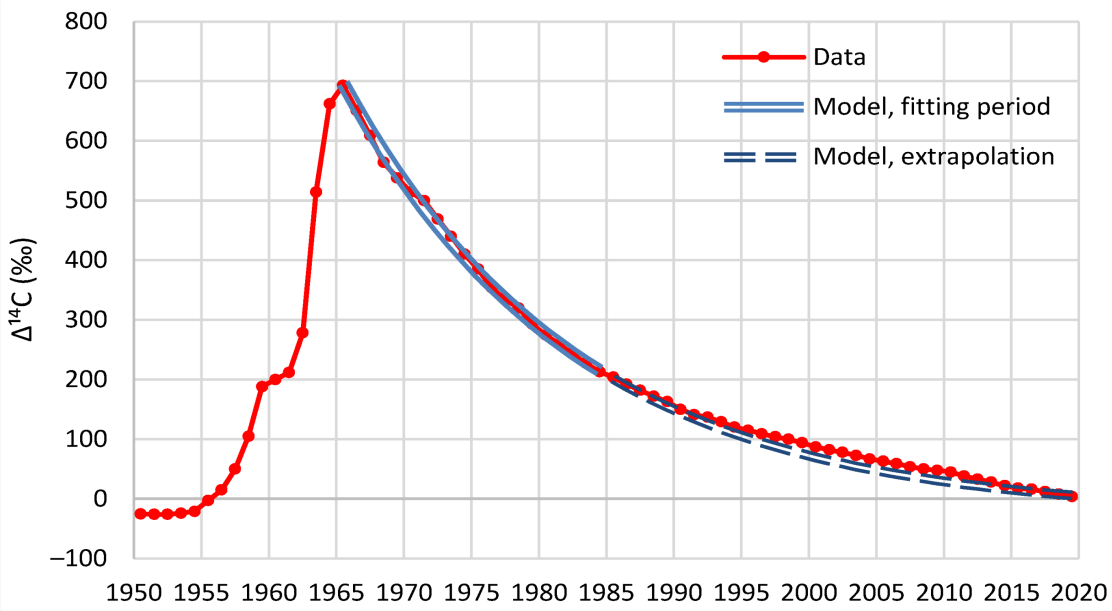


***Figure 2**. Global $\Delta^{14}C$ time series, as provided by Hua et al. (2022) [54], and fitted linear reservoir model by Koutsoyiannis (2024). Reproduced from [77] under CC-BY 4.0 license.*

Koutsoyiannis (2024) [77] Appendix C fits a single-exponential model to a 20-year subset (1965–1985) of the global Hua et al. dataset, yielding an e-folding time of 17.2 years, and shows that this fit extrapolates to the full 54-year record without re-fitting. Koutsoyiannis notes, "The absolute incompatibility of the IPCC [Bern] model with reality, as demonstrated through the $\Delta^{14}C$ data (and the model fitted to them, which is in perfect agreement with the data) is obvious." Because of several related processes that Koutsoyiannis speculated could lead to the slowing of ^{14}C removal, Koutsoyiannis noted the 17.2-year value he calculated as an "upper bound" of the true atmospheric response time of ordinary CO_2. Figure 2 shows that the single exponential reproduces the observed decay data with extremely high fidelity and no measurable deviation over 54 years. By 2019 the AR2, AR3, AR4, and AR5/AR6 Bern IRFs predict 33%, 35%, 42%, and 48% of bomb-^{14}C remaining in the atmosphere, respectively, while the actual observed value is 4.3% and the single-exponential fit ($\tau = 18.389$ yr, $R^2 = 99.825\%$) over the full 1965–2019 record tracks the data within residual noise (see Figure 1). For a comprehensive analysis and discussion of the single-exponential fit of the bomb-^{14}C global data series from Hua et al. (2022) [54], please refer to Cohler & Soon (2026) [23].

The clean exponential decay of the bomb-^{14}C record shows no long tails, and therefore this observed single exponential functional form itself falsifies two central assumptions of the Bern impulse-response function of AR5/AR6. First, the Bern model includes a permanent non-decaying airborne fraction (a_0), which requires the decay curve to approach a non-zero horizontal asymptote at long times (see Figure 1 above). Second, the Bern model is a sum of multiple exponential terms with widely different e-folding times (AR2: 41.2 yr; AR3: 45.6 yr; AR4: 95.6 yr; AR5/AR6: 165 yr), which all differ dramatically from the fitted e-folding times of 17.2 or 18.4 yr, not to mention the clear curvature deviations of the Bern models from the bomb-^{14}C curve (see Figure 1). The actual data and the Bern model curve are therefore mathematically, physically, and visually incompatible.

The clean single-exponential character of the bomb-^{14}C decay therefore constitutes a direct empirical falsification of the Bern models and IPCC's claims of multi-millennial CO_2 persistence. Natural removal processes do not fundamentally distinguish among the CO_2 isotopologues. The only rate-affecting isotope effect at the air-sea interface is the Schmidt-number-driven kinetic correction, approximately 0.7‰ between $^{14}CO_2$ and $^{12}CO_2$. The large biospheric and oceanic isotopic offsets (approximately −18 to −20‰ for C3 photosynthesis; approximately 8‰ for atmosphere-ocean equilibrium) are stationary partition effects on the flux ratios, not separate transport equations with separate eigenvalues. The atmospheric relaxation operator established by the Hua et al. (2022) [54] decay is therefore linear and common across isotopologues to within per-mil accuracy, so the single-exponential e-folding time derived from the ^{14}C label applies to the

atmospheric CO_2 reservoir as a whole at that same accuracy. A "tracer not bulk" interpretation of the bomb-^{14}C decay is incompatible with the linearity the data establish.

4.2 Revelle Factor, Ocean Flux Claims, SOCAT Limitations, and the Global Carbon Budget

The Bern model's [68,70] central reliance on the Revelle factor [104,132] to justify reduced uptake efficiency and the permanent a_0 fraction misapplies a strictly local surface property as a global bulk ocean kinetic barrier. Human emissions and land-use changes, at ~11–12 GtC per year [39], represent only ~5–6% of the natural carbon cycle's gross annual turnover (~200 GtC per year across terrestrial and oceanic exchanges; [62]). The gross bidirectional exchange of CO_2 between the ocean and atmosphere has been estimated in a broad range of 60–100 GtC per year pre-industrially, with substantial swings in successive assessments reflecting large uncertainties (30–50% relative or more) stemming from differing parameterizations of gas transfer velocity, wind fields, and tracer constraints such as bomb-^{14}C inventories ([59] Chapter 6, Figure 6.1; [61] Chapter 5; [126]). These wide-ranging estimates in recent decades highlight large uncertainties and sensitivity to methodological choices.

Regardless of the exact figure within this uncertain range, anthropogenic emissions (~10.3 GtC yr^{-1}) represent only a minor perturbation on the natural gross exchange, highlighting the carbon cycle's overwhelming dominance by rapid bidirectional cycling that has self-regulated for billions of years without permanent disequilibrium. This enormous bidirectional flux, driven by regional differences in temperature and solubility, is sustained and amplified by vigorous ocean circulation operating on multiple timescales.

Following CO_2 in-gassing in cooler high-latitude waters, which elevates dissolved inorganic carbon (DIC) concentrations, this DIC-enriched water is transported equatorward: a smaller portion (10%–25%) travels rapidly along Deep Western Boundary Current pathways on timescales of months to 1–2 years, while the majority (60%–80%) moves via slower interior gyre and overturning circulation pathways over several years [14]. Upon reaching warmer low-latitude regions, the elevated DIC combined with temperature-driven reductions in solubility drives substantial outgassing [29,127]. Critically, only a small portion (~7% globally) of this carbon ever reaches the very deep ocean and remains sequestered for decades or longer [73,106]; the vast majority recirculates in upper and intermediate layers (500–2000 m) and is quickly released back to the atmosphere. This redistribution therefore functions as rapid negative feedback to high-latitude CO_2 in-gassing, counteracting the perturbation on human-relevant scales [29].

4.2.1 Empirical Evidence of Robust Self-Regulation and No Permanent Airborne Fraction

Under current conditions, this self-regulating mechanism results in no detectable slowdown in absorption by ocean and land sinks, as directly evidenced by linearly increasing absolute uptake proportional to atmospheric CO_2 over the instrumental record despite ~50% pCO_2 rise since pre-industrial times [62] while paleoclimate and geological evidence demonstrate complete drawdown from levels during the last 500 million years that reached several thousand ppm without residual permanence.

The IPCC's claimed 2–3 GtC yr^{-1} global net ocean flux, however, remains empirically unconstrained due to fundamental limitations in measurement coverage, resolution, and simultaneity; direct observations are sparse, non-simultaneous, and heavily extrapolated, yielding uncertainties that dwarf any claimed small net value. The system's rapid surface feedbacks and historical resilience to far larger natural perturbations confirm its capacity for proportional rebalancing without requiring irreversible anthropogenic dominance or local Revelle-factor imposed global limitations. Far from a fragile system prone to irreversible saturation, the Earth carbon cycle exhibits robust self-regulation on all observed timescales, with no empirical precedent for a non-decaying airborne fraction as posited in Bern models.

The absence of any detectable marginal slowdown directly falsifies Revelle as a dominant limiter. Instead, the cycle's dominance by natural gross bidirectional fluxes ensures rapid re-equilibration, rendering long-term anthropogenic persistence a modeling artifact rather than physical reality.

4.2.2 The Revelle Factor in Context: Local Effect, Not Global Barrier

The Revelle factor is defined using local atmospheric partial pressure of CO_2 (pCO_2) and local DIC as [33]:

$$\mathrm{R} \equiv \frac{\partial \ln pCO_2}{\partial \ln \mathrm{DIC}} \tag{10}$$

In the surface waters at the three fixed open-ocean time-series stations that provide the longest empirical records (HOT, BATS, ESTOC; [7,8,44,71,107,119]), the local value is typically around 10. This local buffering reduces marginal uptake efficiency at the air-sea interface, as surface pCO_2 rises faster, partially offsetting the influx gradient. Crucially, this is a strictly local, differential effect confined to the immediate surface layer where gas exchange occurs; it does not imply global "saturation" or act as a planetary barrier. Direct multi-decadal empirical evidence of Revelle changes exists at only these three single fixed geographic stations, each covering effectively 0% of the global ocean surface area (~361 million km²). These three isolated points are the longest and highest-quality open-ocean records available worldwide; no comparable long-term empirical data exist in colder subpolar, polar, or major upwelling zones. The full SOCAT database (~41 million fCO_2 observations spread non-synoptically over 68+ years) provides coverage of less than 1–3% of the global ocean surface area in any given year and is heavily biased toward Northern Hemisphere shipping routes. Any purported "global" Revelle trend therefore requires precisely the same sparse, route-biased, non-synoptic sampling followed by extensive interpolation, extrapolation, gridding, and area-averaging of an intensive property across heterogeneous non-equilibrium water masses as detailed below (see Section 4.2.4) to render the result: a statistical artifact lacking thermodynamic physical meaning (see also [20,21,37]).

The ocean constitutes a vast, circulating reservoir with a volume of 1.33×10^9 km³ and a DIC inventory of approximately 38,000 GtC. Circulation continually refreshes surface waters with deep, undersaturated DIC via upwelling, while surface currents transport excess surface DIC to warmer waters, and downwelling thermohaline circulations transport it to the deeper ocean. This dynamic process strongly suggests that the global ocean remains far from saturation on all policy-relevant timescales, allowing net global in-gassing and out-gassing to proceed unimpeded by local Revelle factors, whether or not there is any net imbalance between the atmosphere and ocean.

4.2.3 Realistic Emission Scenarios, Biospheric Benefits, and Ultimate Removal

Although this paper demonstrates, through refined reservoir routing [77] and mass-balance analysis, that natural biogeochemical fluxes dominate the carbon cycle and that anthropogenic emissions produce no detectable long-term accumulation in the atmosphere (atmospheric residence time ~3.5–4 years), the following calculations are presented solely for illustrative purposes under conventional mainstream assumptions that all recoverable fossil carbon would enter the atmosphere and require absorption by natural sinks. These scenarios therefore do not endorse that assumption; they merely show the ocean's and biosphere's capacity even under the most extreme consensus-based emission inventories.

Quantitative mass-balance calculations illustrate this capacity even under extreme scenarios. Consider an upper-bound estimate of all recoverable fossil carbon (~5,000 GtC). For a theoretical instantaneous pulse, the buffered equilibrium approximation $\mathrm{DIC} \approx \mathrm{DIC}_0 \times (pCO_2/pCO_2{}^0)^{1/\mathrm{R}}$ with $R = 10$ and $pCO_2{}^0 = 280\mathrm{ppm}$, yields a final $pCO_2 \approx 750 - 780\mathrm{ppm}$, with the ocean absorbing ~80–84% of the added carbon ($\Delta\mathrm{DIC}_{\mathrm{ocean}} \approx$ 4,000–4,200 GtC, or just an ~11% increase in inventory) and full equilibration occurring without unabsorbed residue.

In a more realistic scenario, still assuming the conventional human attribution solely for illustration, these emissions would occur gradually over centuries rather than instantaneously. Current anthropogenic carbon emissions are approximately ~10.3 GtC per year [39]. Assuming no major transitions to alternative energy sources and modest growth in demand, exhaustion of 5,000 GtC could span 300–500 years; more conservative projections accounting for efficiency gains or partial 'decarbonization' extend this to 400–600 years or longer. During this prolonged period, ocean uptake proceeds continuously and proportionally to the evolving pCO_2 local gradients, absorbing a substantial fraction (~44–56%) of annual emissions concurrently. This concurrent absorption prevents the full cumulative 'excess' from accumulating in the atmosphere, resulting in lower

peak concentrations of ~550–700 ppm compared to the pulse scenario. The system tracks toward equilibrium progressively, with significant redistribution to intermediate/deep waters on centennial scales via thermocline ventilation and near-complete atmosphere-ocean balance within 500–1,000 years.

A gradual rise to stabilized levels of 550–700 ppm over centuries would confer substantial biospheric benefits. Extensive evidence from free-air CO_2 enrichment experiments and satellite observations documents enhanced net primary productivity (~20–30% global increase already from ~280 to 420 ppm [47,86,134]), greening of arid regions, improved water-use efficiency, and higher crop yields under elevated CO_2. Optimal photosynthetic rates for C3 plants (most crops/forests) occur at 800–1,200 ppm, with no direct physiological toxicity to humans or mammals below ~5,000–10,000 ppm. Paleoclimate records show thriving ecosystems at 1,000–2,000 ppm, supporting the view that such levels enhance food security and ecological resilience.

Ultimately, geological sinks (primarily silicate weathering and sedimentation) ensure complete removal of 'excess' carbon on 10,000–100,000-year scales, as Earth's history repeatedly demonstrates full drawdown from levels of several thousand ppm without any permanent airborne residue [9].

4.2.4 Severe Limitations of SOCAT Data for Claimed Global Net Ocean CO_2 Flux

Moreover, precise global net CO_2 flux between atmosphere and ocean, both current (~2.5 GtC yr^{-1} claimed sink) and future, remains empirically unconstrained due to fundamental measurement limitations. Direct air-sea flux relies on sparse, non-synoptic pCO_2 observations (e.g., SOCAT database) covering a tiny fraction of ocean space-time effectively. Global estimates require massive extrapolation assuming smoothness, introducing uncertainties from aliasing variability and sub-grid-scale processes that are on the order of the claimed net signal [30,45,53].

SOCAT compiles surface ocean fugacity of CO_2 (fCO_2) measurements, primarily from voluntary observing ships and research cruises, with minor contributions from moorings, drifters, and emerging biogeochemical Argo floats. Although the database has accumulated tens of millions of observations over seven decades, direct coverage in any given year amounts to less than 1–3% of the global ocean surface area (~361 million km^2), leaving >97–99% unsampled [4,105]. Sampling is highly irregular and biased, concentrated along major Northern Hemisphere shipping routes (e.g., North Atlantic and North Pacific trade lanes), with systematic undersampling of the Southern Ocean, high latitudes, remote open-ocean regions, and coastal zones during extreme seasons or weather [38,53]. No synoptic global coverage exists, and many areas remain unobserved for years to even decades.

Global gridded fCO_2 fields necessary for bulk flux calculations are generated through statistical interpolation or machine-learning extrapolation across these vast data voids [45,83]. These methods employ covariate proxies, principally sea surface temperature (SST) and satellite-derived chlorophyll-*a* concentration, based on thermodynamic (solubility) and biological (photosynthetic drawdown) relationships observed in the limited sampled regions. However, satellite SST retrievals measure radiative skin temperature (top ~10–20 μm layer; e.g., as detailed in GHRSST [Group for High Resolution Sea Surface Temperature] documentation [31,97]), not bulk temperature (0.1–5 m depth), and require calibration against a sparse in situ network comprising approximately 4,000 Argo floats, 1,500–2,000 drifting buoys, and several hundred moored platforms, resulting in a global measurement density of roughly one point per 200–300 km [3]. Chlorophyll proxies are similarly indirect and often unreliable in high-nutrient low-chlorophyll regimes or where non-phytoplankton processes dominate [45].

These extrapolated products are frequently described as "observation-based" or "data-driven" in syntheses such as the Global Carbon Budget to differentiate them from fully process-based models [39]. This terminology, however, substantially overstates the empirical grounding, as the global net flux integral is predominantly determined by assumptions embedded in the extrapolation rather than direct measurement across the ocean domain [53]. Ensemble spreads among independent mapping approaches highlight sensitivities to proxy choices and methodological differences, with regional uncertainties, especially in the Southern Ocean, commonly equaling or exceeding the estimated mean flux [30,43].

These limitations extend directly to high-level assessments such as IPCC AR6 [62] (Figure 5.12), where the central anthropogenic net ocean sink (2.5 ± 0.6 GtC yr^{-1}) is reproduced from Global Carbon Budget syntheses that incorporate these same extrapolation-heavy pCO_2 products alongside process-based models. The apparent precision of this policy-relevant number thus inherits the profound uncertainties from SOCAT's observational sparsity, amplifying concerns over the representation of the global carbon cycle as empirically robust when it remains overwhelmingly inference-driven.

4.2.5 Model-Derived Global Carbon (CO_2) Budget

IPCC AR6 Working Group I Figure 5.12 presents a schematic of the global carbon budget for 2010–2019, partitioning air-sea CO_2 fluxes into pre-industrial "natural" components (yellow arrows, ~54 GtC yr^{-1} gross each way) and anthropogenic "perturbation" components (pink arrows, with gross influx ~25.5 GtC yr^{-1}, outflux ~23 GtC yr^{-1}, yielding a net anthropogenic sink of 2.5 ± 0.6 GtC yr^{-1}). The caption cites a diverse array of sources for reservoirs and fluxes, including rescalings from historical estimates [108], tracer-based adjustments (bomb-^{14}C inventories), and multiple model outputs (e.g., dynamic global vegetation models for land photosynthesis; ocean biogeochemistry simulations for export production, remineralization, and deep reservoirs; extrapolations for anthropogenic carbon inventories from [46]).

Critically, the much-quoted IPCC AR6 headline net anthropogenic ocean sink of 2.5 ± 0.6 GtC yr^{-1} is reproduced directly from Friedlingstein et al. (2020) [40], the Global Carbon Budget 2020 assessment. This value emerges from a combination of two ensembles: (i) nine global ocean biogeochemistry models (GOBMs) sharing common forcings, parameterizations, and structural assumptions; and (ii) four pCO_2-based mapping products derived from SOCAT data via heavy extrapolation (as detailed above). The ±0.6 GtC yr^{-1} uncertainty is reported as the standard deviation across this non-independent ensemble. This approach implicitly relies on the logic of the Central Limit Theorem (CLT). That theorem states that averaging multiple independent estimates with random errors should produce a mean closer to the true value, with uncertainty scaling as the standard deviation. However, the approach applies this logic without any qualification. None of the CLT's required conditions are satisfied. The models are not independent: they share common forcings, architectures, and calibrations. Their errors are not randomly distributed but are structurally correlated and systematically biased. Finally, the sample size of only nine estimates is far too small for any meaningful convergence. The resulting spread thus measures internal modeling disagreement, not probabilistic uncertainty around a true value, systematically underestimating the far larger epistemic errors arising from shared limitations and observational sparsity [40,53].

The gross partitioned fluxes in AR6 Figure 5.12 (e.g., anthropogenic in/out arrows) lack any empirical source, as CO_2 molecules cannot be separated by origin in real-time exchange. They arise exclusively from model simulations comparing perturbed and pre-industrial counterfactual runs, with pre-industrial gross values further adjusted via indirect tracer rescaling rather than contemporary measurement. The figure caption omits explicit acknowledgment that nearly all arrows and numbers, particularly the gross anthropogenic components, are pure model constructs with no observable basis, presenting a stylized partition that conveys precision unattainable from available data.

4.3 Millennial Persistence as a Modeling Artifact

The Bern model's non-decaying term and millennial-scale tails constitute parameterization artifacts that artificially constrain ocean mixing and elevate the local Revelle buffering effect into an unfounded global ocean bulk mass kinetic barrier. No physical mechanism or empirical data supports this transformation: Revelle remains a differential surface property incapable of overriding the cycle's rapid bidirectional feedbacks and self-regulating dominance. Assertions of inevitable centuries-to-millennia anthropogenic persistence are thus detached extrapolations, incompatible with mass conservation, the absence of any detectable human isotopic or dynamic fingerprint, and the carbon cycle's unbroken record of resilience across all timescales: from the stable net input signature since the Little Ice Age and throughout the instrumental era, through the remarkable constancy of the previous 2,000 years and broader Holocene, the full glacial-interglacial swings of the past 100,000–800,000 years, the dramatic excursions over millions of years, to the

complete re-equilibrations from extreme levels throughout the 500 million years of the Phanerozoic. The Bern framework perpetuates an illusion of unique, quasi-permanent human disruption where none is required, or evidenced, by the data.

5. From Bern Model to CMIP6: Modeling Shifts and Thermodynamic Invalidity of GMST

5.1 IPCC AR6 Shift from Bern Model to CMIP6 Earth System Models for Atmospheric CO_2

In IPCC AR6, the primary assessment of the carbon cycle in Chapter 5 [62] abandoned reliance on a Bern-derived multi-exponential impulse response parameterization with fixed coefficients, instead favoring ensemble outputs from CMIP6 of the Coupled Model Intercomparison Project (CMIP) ESMs that incorporate ensemble spread, state-dependent nonlinearities, and qualitatively similar long-term CO_2 persistence (~20–30% remaining airborne over millennia). Chapter 5 justifies this shift as an "advancement," stating: "A larger number of Earth System Models (ESMs) from CMIP6 compared to CMIP5 are used to assess the contemporary carbon cycle and its historical evolution" and emphasizing "improved representation of key processes" such as nutrient limitations, permafrost thaw, and asymmetries in emissions/removals [62] (AR6, Section 5.1). No acknowledgment is made of the Bern model's physical flaws or the superior empirical fits of linear alternatives; instead, the move to ESMs is framed as progress toward capturing "emerging complexities" absent in reduced-form models. Note, however, that for 'simplified' climate metric calculations, Chapter 7 retains the AR5 Bern model from Joos et al., 2013 [68], to maintain consistency in Global Warming Potential [GWP] and Absolute Global Warming Potential [AGWP] values between AR5 and AR6.

This progression from transparent reduced-form equations to opaque, high-dimensional ESMs tuned to thermodynamically invalid GMST represents not scientific advancement but a continuation of unresolved limitations. The term "Earth System Model" (ESM) itself, as applied to CMIP6 models, introduces a distinction that overstates differences from prior Atmosphere-Ocean General Circulation Models (AOGCMs/GCMs). At their core, ESMs retain the identical CMIP model dynamical framework, including discretized Navier-Stokes equations for fluid motion, radiative transfer schemes, sub-grid parameterizations, and fully coupled physical components. The primary additions consist of supplementary biogeochemical modules (e.g., interactive carbon cycles, nutrient constraints, and dynamic vegetation), which do not alter the underlying physical core. These extensions still fail to address the foundational thermodynamic invalidity arising from calibration to physically meaningless GMST, an arbitrary statistical construct without a unique physical definition [20,37]. The reclassification under "Earth System" nomenclature, accompanied by assertions of greater realism, shifts emphasis toward increased complexity without resolving core limitations, thereby reducing transparency and complicating direct comparison with simpler, analytically solvable alternatives grounded in first-principles physics and direct observational fidelity. All versions rely on unverifiable assumptions of parallel vs. serial sinks and molecular discrimination, extrapolated far beyond any empirical validation window (~170 years of reliable data), with zero direct observational evidence for long tails, permanent fractions, or selective saturation. Such evolution underscores a reliance on complexity that diverges from alignment with verifiable physics.

5.2 First-Principles Proof of the Thermodynamic Invalidity of GMST

A foundational critique of contemporary climate metrics arises from rigorous thermodynamic principles, which demonstrate that no physically meaningful global temperature can exist for a system as heterogeneous and non-equilibrium as the Earth's surface-air-ocean interface. Essex et al. (2007) [37] provide a comprehensive mathematical proof grounded in classical thermodynamics, emphasizing that temperature is an intensive property, defined locally in equilibrium subsystems through the relation, The relevance of GMST to CO_2 attribution is direct and not peripheral: every CMIP6 Earth System Model used by AR6 Chapter 5 to replace the explicit Bern IRF is tuned to reproduce historical GMST, so any ESM-derived airborne-fraction or ocean-sink number inherits whatever physical status GMST itself possesses.

$$T = \left(\frac{\partial U}{\partial S}\right)_{V,N} \tag{11}$$

where U is internal energy, S entropy, V volume, and N particle number ([82], Ch. I, §3; [15], Ch. 1, pp. 15–17). Intensive properties, by definition, do not scale with system size and cannot be meaningfully summed or averaged across subsystems without reference to additive extensive quantities. They are qualities of an equilibrium system such as temperature, pressure, density, pH, or Revelle factor. The Earth's climate system, encompassing vast disparities in mass density (air versus seawater differing by over 800-fold), heat capacities (oceans holding thousands of times more thermal energy than the atmosphere per unit volume), and dynamic non-equilibrium states (driven by solar gradients, convection, and phase changes), precludes the existence of a single, thermodynamically valid global temperature. Averaging local temperatures, whether arithmetic, geometric, or otherwise, produces statistical constructs lacking physical uniqueness, as infinite families of means, e.g. Hölder means

$$A_r = \left(\frac{1}{n}\sum_{i=1}^{n} x_i^r\right)^{1/r} \tag{12}$$

can be applied to the same dataset, yielding divergent numerical values and even opposite trends over time. No physical law prescribes one averaging rule over another for temperature in such systems; gradients, not averages, govern heat flows and dynamics per the second law of thermodynamics. Thus, any claimed "global temperature" is an arbitrary artifact devoid of thermodynamic meaning, incapable of representing the system's energy state or changes therein.

Cohler (2025) [20] builds upon and extends this foundational analysis, directly confronting the IPCC's GMST, which is an area-weighted average of near-surface air temperatures over land and sea ice with sea surface temperatures over open water. This construction fundamentally violates thermodynamic principles by treating intensive temperatures as if they were extensive, ignoring the massive differences in mass and specific heat capacity required for true energy calculations, such as $Q = mc\Delta T$, where changes in thermal energy depend on these extensive properties.

Attaching units of degrees Celsius (°C) to such statistical averages, which are not themselves physical temperatures, misrepresents them as genuine thermodynamic temperatures. This practice conflates a relative index derived from temperature anomalies with an actual thermodynamic state variable. Different averaging methods such as arithmetic (common in IPCC datasets), harmonic (relevant for certain resistances or rates), root mean square (tied to kinetic energy), or higher powers (e.g., fourth root for blackbody radiation scaling with T^4), produce substantially different results and trends from identical measurements, underscoring the arbitrariness absent any guiding physical principle.

Furthermore, IPCC glossary definitions of GMST and global warming are circular. Datasets combine incomparable measurement types across eras (e.g., bucket versus engine intake sea temperatures, sparse historical coverage), and extensive homogenization and other "adjustments" often generate trends comparable in magnitude to the claimed signals themselves [22,24]. Critically, all CMIP models are explicitly tuned to reproduce historical GMST trajectories, thereby inheriting GMST thermodynamic invalidity; as coupled systems, errors in CMIP models propagate across all variables, rendering projections scientifically invalid [20,22]. Cohler (2025) [20] notes that this systematic misrepresentation elevates a statistical construct without physical referent to the core of climate science, making GMST-based claims such as warming magnitude, human attribution, future projections, and associated impacts, thermodynamically impossible and unsupported by first principles.

6. Causality Between Local Temperatures and CO_2

Determining the directional relationship between local atmospheric temperatures and CO_2 concentration is crucial for understanding the drivers of observed changes in the Earth's climate system. High-resolution local proxy records and direct observations from every major archive examined demonstrate unidirectional temperature → CO_2 causality on all timescales, with no empirical evidence anywhere showing the reverse.

Ice cores from Antarctica (Vostok, EPICA Dome C, WAIS Divide, Taylor Dome) and Greenland (GISP2) provide the clearest high-resolution evidence. In every glacial termination and every millennial-scale event over the past 800,000 years, local temperature consistently leads CO_2 by 200 to over 1,000 years, with no exceptions [42,92,100]. A comprehensive review of the paleoclimate literature covering the past 650,000 years reaches the same conclusion: atmospheric CO_2 variations generally follow changes in temperature and other climatic variables rather than preceding them [115]. No quantitative validation exists for CO_2 as the prime mover in glacial-interglacial cycles.

This unidirectional pattern extends throughout the Phanerozoic eon. Reconstructions spanning the last 500 million years, pairing $\delta^{18}O$-based local temperature proxies with independent CO_2 proxies (boron isotopes in foraminifera, stomatal density in fossil leaves, and alkenone $\delta^{13}C$), show temperature changes preceding CO_2 variations with characteristic lags on the order of millions of years [79]. Modern observations reinforce the same causality on shorter timescales. Major volcanic eruptions (Pinatubo 1991, El Chichón 1982, Agung 1963) cause rapid aerosol-driven surface cooling followed by measurable slowdowns or temporary drops in atmospheric CO_2 growth rate [41]. El Niño warming events precede global CO_2 growth-rate spikes by approximately six months [28,84,103]. Satellite (OCO-2) and ground-based eddy-covariance (FLUXNET) measurements show regional CO_2 fluxes directly tracking local temperature gradients and anomalies, with warmer regions acting as net sources and cooler regions as net sinks [5,35]. Recovery sequences after volcanic, ENSO, and warming events consistently show temperature recovery preceding CO_2 recovery.

The physical mechanisms underlying this precedence are relatively well understood and temperature-dependent: higher temperatures accelerate microbial respiration in soils and vegetation according to the Q_{10} rule (typically $Q_{10} \approx 2–3$), releasing CO_2, while warming reduces CO_2 solubility in seawater, driving oceanic outgassing. We demonstrate that these natural processes can account for observed CO_2 increases on instrumental timescales without requiring detectable anthropogenic forcing, as human emissions represent only a minor perturbation rapidly integrated into the temperature-modulated cycle. Reverse causality ($CO_2 \rightarrow$ temperature) is unsupported by any empirical record. Temporal precedence rules out CO_2 as the initiating driver on every observed timescale; the independent isotopic test, the Keeling net-input signature stationary at −13.2‰ for 500+ years, further bounds any $CO_2 \rightarrow T$ feedback contribution via fossil-fuel accumulation to a magnitude below the detection limit of $\delta^{13}C_I$ [78].

7. Keeling Plots: Evidence for Overwhelming Biosphere Dominance

The isotopic composition of atmospheric CO_2, particularly the ratio of ^{13}C to ^{12}C expressed as $\delta^{13}C$, offers a powerful diagnostic tool for distinguishing between natural and anthropogenic sources of CO_2. Fossil fuel emissions are strongly depleted in ^{13}C (typically around −28‰ due to preferential uptake of lighter ^{12}C over ^{13}C during photosynthesis), so a significant human contribution to rising atmospheric CO_2 should manifest as a detectable shift in the net isotopic signature of sources and sinks (i.e. the isotopic input signature $\delta^{13}C_I$) toward more negative values. However, detailed analyses of both instrumental and proxy records reveal no such shift, pointing instead to overwhelming dominance by natural biospheric processes. Recent evidence also clearly demonstrates enhanced biospheric net greening both on land [47,86,134] and ocean surface [113,133].

Central to this evidence are Keeling plots, which graph $\delta^{13}C$ against the inverse of CO_2 concentration ($1/[CO_2]$) (see Figure 3). In a system with constant net input signature, these plots yield straight lines where the y-intercept represents the average isotopic composition of sources minus sinks, $\delta^{13}C_I$. Mass-balance derivations show that for constant inputs, $\delta^{13}C = \delta^{13}C_I + (\delta^{13}C_0 - \delta^{13}C_I)([CO_2]_0 / [CO_2])$, producing perfect linearity; deviations indicate varying signatures. Across four instrumental sites including Mauna Loa, Barrow, La Jolla, and the South Pole, and spanning over 40 years (Figure 3), these plots exhibit extraordinary linearity with correlation coefficients R^2 up to 0.99, and intercepts consistently clustering between −12.9‰ and −13.3‰, averaging approximately −13.2‰ globally. Subperiod analyses, moving windows, and trend tests reveal no systematic decrease in this intercept despite anthropogenic emissions more than doubling during the 40-

year period; any local trends are small (±0.01 to ±0.38‰ per decade), often positive, and statistically insignificant [78].

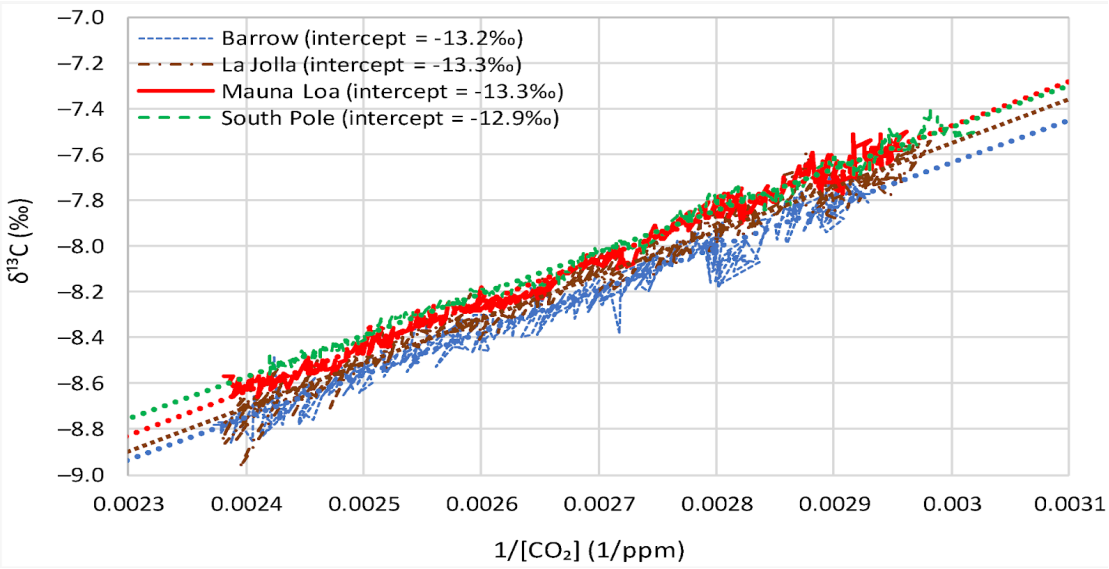


Figure 3. *Keeling plots of seasonally adjusted $\delta^{13}C$ data from four instrumental sites (Barrow, La Jolla, Mauna Loa, South Pole), demonstrating high linearity and stable net input signatures (intercepts −12.9‰ to −13.3‰) with no discernible trend despite rising atmospheric CO_2. Reproduced from [78] under CC-BY 4.0 license.*

Proxy data from coralline sponges and ice cores/firn air, digitized and spanning ~1520–1997 AD (covering the transition from Little Ice Age to modern), extend this finding: Keeling plots remain linear with an intercept of ~ −13.2‰, identical to instrumental values, and subperiod intercepts show no consistent decline aligned with rising human emissions, instead fluctuating without directional bias. Seasonal variations, strongest in northern high-latitude sites due to terrestrial biosphere influence, arise from photosynthesis preferentially removing lighter ^{12}C (enriching atmospheric $\delta^{13}C$ during growth seasons) and respiration releasing depleted CO_2 (depleting it during decline phases). Yet these seasonal effects scatter around the over-annual linear trend without altering the stable net signature.

A simple deterministic model incorporating only biosphere-driven seasonal fractionation (global photosynthesis at ~ −27.6‰ during CO_2 decline phases and site-varying respiration −13 to −25‰) reproduces both time series and Keeling plots with explained variances of 97.8–99.5% for instrumental data and 95.9–98.4% for proxies. Critically, no term for anthropogenic emissions is included or needed; adding one does not improve fits. The observed long-term $\delta^{13}C$ decline (the so-called "Suess effect") and parallel CO_2 rise thus emerge from temperature-amplified imbalances in natural fluxes, biosphere expansion increasing respiration and release of depleted CO_2, overwhelming any potential fossil-fuel signal. Natural reservoirs (e.g., C3 plants, soil organic matter) overlap or exceed fossil depletion ranges, and with fluxes >20 times larger than human inputs, anthropogenic CO_2 blends indistinguishably into the dominant biosphere signature ([22,78]; Fig. 2 in [80]). This isotopic stability provides independent, compelling evidence that natural processes fully drive recent atmospheric CO_2 changes.[2]

2. Following publication of Koutsoyiannis [78], Kleber [74] raised several objections to the net isotopic signature findings. In a subsequent peer-reviewed Reply [80], Koutsoyiannis addressed every substantive point raised by Kleber in a direct, line-by-line response. The Reply demonstrates that (i) photosynthesis seasonally increases atmospheric $\delta^{13}C$ (exactly as stated and shown graphically in the original analysis), (ii) biosphere respiration releases CO_2 even more depleted in ^{13}C than fossil fuels, rendering the ~4% anthropogenic contribution undetectable against the 96% natural flux, (iii) Kleber's attempted falsification using the Eggleston et al. (2016) [32] paleorecord yields only negligible overall correlation ($r \approx 0.07$) with alternating signs across optimized segments, and (iv) temperature–$\delta^{13}C$ relationships alternate in sign over glacial cycles, all directly contradicting claims of a consistent

8. Top-of-Atmosphere Radiative Imbalance as the Potential Driver of Observed Atmospheric CO_2 Rise Through Ocean and Soil Outgassing

Net TOA flux equals incoming solar irradiance minus reflected shortwave minus emitted longwave radiation. NASA CERES measures broadband shortwave radiances (SW, 0.3 to 5 μm) and total radiance (TOT, 0.3 to 200 μm) ([88], p. 896) among other bands. Broadband longwave LW, not measured directly, is derived as TOT minus SW during daytime and TOT alone at night [88], p. 897. Any SW calibration error therefore propagates into the derived daytime LW.

The unadjusted CERES EBAF Edition 4.0 global-mean net TOA imbalance for July 2005 through June 2015 is 4.3 W m^{-2} [88], Table 5. The reported EBAF value of 0.71 ± 0.10 W m^{-2} at 95% CI [88], Section 2b(3) is *not* a CERES satellite measurement. It is the sum of three in situ ocean and cryosphere heat uptake purported "estimates," with CERES SW and LW fluxes shifted by a total of 3.5 W m^{-2} but with global net shifted by 3.59 (*not 3.5*) W m^{-2} under an "objective constrainment algorithm" to match that sum.

Loeb et al. (2018) [88] does not state a comprehensive global 1σ uncertainty on the net TOA flux anywhere in the paper. The paper does, however, claim (p. 904) that the 3.5 W m^{-2} "shift" is "within [the SW and LW] ranges of uncertainty." Under the standard metrology convention that "within uncertainty" means within the standard uncertainty (1σ), this claim requires global 1σ ≥ 3.5 W m^{-2}. By the logic in [88], their global uncertainty must be at least as large as the shift they made.

The ±0.05 W m^{-2} 1σ uncertainty on the EBAF value comes from the Argo-derived target, not from NASA CERES, and is at least 70 times tighter than the CERES global 1σ implied by [88]'s own constrainment claim. Cohler et al. (2026) [21] shows that the Argo target is itself physically invalid; even accepting its construction arguendo, its own uncertainty exceeds ±1 W m^{-2} at 95% CI, making the IPCC EEI value statistically indistinguishable from zero. Loeb et al. (2020) [90] confirms the construction by applying "an offset to the satellite EEI such that its mean value over the 15-year period" matches the in situ ocean mean, and acknowledging that satellite fluxes are "not at the level of accuracy required" to resolve EEI absolutely.

Against the [88]-implied CERES global 1σ of ≥ 3.5 W m^{-2}, the reported 0.71 W m^{-2} is at most 0.20σ above zero, below any standard detection threshold. The unadjusted CERES measurement therefore carries a 1σ envelope of approximately 4.3 ± 3.5 W m^{-2} by [88]'s own logic, but 4.3 – 0.71 = 3.59 W m^{-2} not 3.5 W m^{-2}. We explain this further serious discrepancy in Appendix A which documents the citation chain, the uncertainty-budget manipulations, the inference of the global 1σ that [88] failed to state, and the σ-count collapse of the detection claim.

Within this large uncertainty envelope, a value of ~2.6 W m^{-2} represents the best-constrained natural imbalance that quantitatively accounts for the entire observed rise in atmospheric CO_2 concentration from 2000 to 2025 through natural biogeochemical processes. When combined with van 't Hoff solubility kinetics and Arrhenius respiration kinetics, it quantitatively reproduces the observed atmospheric CO_2 growth rate (~2.25 ppm yr^{-1} average from Scripps/NOAA data) with high fidelity using only unadjusted CERES measurements and fundamental physical laws.

A net TOA flux imbalance of 2.6 W m^{-2} (denoted here as F) corresponds to a global annual energy gain of ~41.9 ZJ yr^{-1} ($F \times A \times t$, where $A \approx 5.1 \times 10^{14}$ m^2 is Earth's surface area and $t = 3.156 \times 10^7$ s is one year). Applying standard energy partitioning (approximately 91% to the oceans, ~6% to the land surface, with the remaining ~3% going into cryosphere melt and atmospheric warming), this yields ~38.2 ZJ yr^{-1} of oceanic energy uptake. Using van 't Hoff solubility kinetics ($\partial \ln pCO_2 / \partial T \approx 0.0423$ K^{-1}; [109,125,129]) applied to the upper mixed layer produces an outgassing flux of ~2.44 GtC yr^{-1}. Concurrently, the land-surface component (~2.5 ZJ yr^{-1}) drives enhanced soil and terrestrial respiration via Arrhenius kinetics ($Q_{10} \approx 1.5$, yielding $d \ln R / dT \approx 0.0405$ K^{-1} [13,27,87] on a baseline flux of ~73 GtC yr^{-1}), contributing an additional ~2.34 GtC yr^{-1}. This combined natural flux of ~4.8 GtC yr^{-1} matches the observed atmospheric growth rate of ~2.25 ppm yr^{-1} (~4.8 GtC yr^{-1}) with high fidelity.

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anthropogenic fingerprint. The original data, linearity, −13.2‰ net signature, and biosphere-dominance conclusion therefore remain fully intact [78,80].

Our proposed 2.6 W m^{-2} radiative imbalance arises entirely from four natural mechanisms operating on decadal to multidecadal timescales:

(a) **Reduction in planetary albedo**, primarily through sustained multidecadal decreases in low-cloud cover, has increased absorbed solar radiation by 1.2–1.7 W m^{-2} since 2000 [88,89].

(b) **Variability in total solar irradiance (TSI)**, including both direct changes and indirect amplification through UV-driven stratospheric and tropospheric circulation adjustments, contributes 0.1–0.5 W m^{-2} [25,26,117].

(c) **Recovery from major volcanic activity and natural aerosol fluctuations** (notably the post-Pinatubo rebound and variations in wildfire smoke and mineral dust aerosols) provide an additional 0.3–0.5 W m^{-2} of positive forcing [110,128].

(d) **Internal ocean-atmosphere oscillations** (ENSO, PDO, AMO and related modes) modulate cloud regimes, sea-surface temperatures, and meridional heat transport, contributing 0.4–0.8 W m^{-2} on decadal timescales [99].

These four components, when constrained by the requirement that the resulting energy input reproduces the observed average atmospheric CO_2 growth rate of ~2.25 ppm yr^{-1} (Scripps/NOAA data, 2000–2025) through van 't Hoff solubility and Arrhenius respiration kinetics, yield a best-estimate net imbalance of ~2.6 W m^{-2}. This best-constrained natural value is obtained without any anthropogenic contribution as required by the stable −13.2‰ Keeling net-input signature, which shows that anthropogenic emissions make no detectable net contribution to the observed atmospheric CO_2 rise.

This analysis exemplifies the proper scientific use of "constraint." The best-constrained natural imbalance of ~2.6 W m^{-2} is obtained solely from unadjusted CERES TOA flux measurements (4.3 ± 3.5 W m^{-2}) and fundamental physical laws (van 't Hoff solubility and Arrhenius respiration). These laws directly dictate the energy required to produce the observed outgassing from ocean and soil, calibrated precisely to the actual measured atmospheric CO_2 growth rate of 2.25 ppm yr^{-1} (Scripps/NOAA average, 2000–2025), with no anthropogenic term required and consistent with the stable −13.2‰ Keeling net-input signature showing that fossil-fuel emissions make no detectable net contribution to the observed rise.

In stark contrast, IPCC AR6 [61] (Fig. 7.2 and Chapter 7) and related works [18,111] push the CERES TOA fluxes to the extreme edges of their uncertainty ranges, with incoming solar radiation pushed to its 90% confidence interval (CI) *minimum* of 340 W m^{-2} while reflected solar is pushed to its *maximum* of 100 W m^{-2} in order to force a net displayed imbalance of 1 W m^{-2} (when combined with 239 W m^{-2} of outgoing IR) to plausibly match the small value (0.7±0.2 W m^{-2}) derived from physically invalid Argo-based OHC estimates (uncertainties exceeding ±1 W m^{-2} at 95% confidence due to mesoscale aliasing, deep-ocean ignorance, polar undersampling, Eulerian-Lagrangian discrepancies, and arbitrary positional assignment). The entire procedure is fully circular: one uncertain dataset is tuned to another, both ultimately calibrated to the physically meaningless GMST metric [21]. Our constraint in this paper is non-circular, grounded solely in verifiable physics and unadjusted observations with both constrained choices well inside their respective uncertainty ranges.

This framework provides an internally consistent, first-principles account of the current state of the atmosphere, oceans, and climate system based on empirical data and fundamental physical laws. This 2.6 W m^{-2} value sits well inside the 4.3 ± 3.5 W m^{-2} (1σ) raw CERES envelope quoted above [88,90]. The parallel energy-balance derivation in Cohler et al. (2026) [21] derives EEI uncertainty from the Argo ocean heat content side and shows that the IPCC AR6 0.7 ± 0.2 W m^{-2} Argo-based value is statistically indistinguishable from zero at the 95% CI. The 2.6 W m^{-2} value is deduced, not tuned: the observed 2.25 ppm/yr growth rate, the measured $\delta^{13}C_I \approx -13.2‰$ (which forces the fossil fraction to approximately zero), the established constants of van 't Hoff solubility (0.0423 K^{-1}) and $Q_{10} \approx 1.5$ Arrhenius respiration (0.0405 K^{-1}), Earth's surface area, and the measured baseline respiration flux of ~73 GtC/yr together uniquely determine the required net TOA flux through conservation of mass. The CERES envelope and the mechanism sum (a)–(d) provide two independent consistency checks on the deduced value, not inputs to it.

9. Discussion

9.1 Relationship to Mainstream Carbon (CO_2) Cycle Understanding

The findings presented here stand in direct tension with the prevailing views as articulated in IPCC assessments. This tension warrants careful examination rather than dismissal, as the methodological approaches employed differ fundamentally in their epistemic commitments. Mainstream carbon cycle science relies heavily on complex ESMs calibrated to reproduce observed trajectories, whereas this analysis prioritizes analytically solvable models constrained by mass conservation and tested against data through split-sample validation.

The central interpretive question concerns what constitutes adequate explanation. The linear reservoir routing approach achieves explained variances exceeding 99.7% for CO_2 concentration dynamics using a single tunable parameter, while the Bern multi-compartment models require numerous coefficients and timescales that have varied substantially across IPCC assessment cycles. By conventional model selection criteria favoring parsimony, the simpler framework demonstrably outperforms. Yet parsimony alone cannot adjudicate between competing physical interpretations; the question is whether the additional complexity of the Bern model captures real physical processes or introduces artifacts.

9.2 Physical Residence Time versus Modeled Adjustment Time

Defenders of mainstream carbon cycle models argue that short residence times and long adjustment times are compatible because they purportedly measure different quantities: residence time reflects mean molecular turnover, while adjustment time characterizes system response to hypothetical perturbations pulses [61]. This distinction, however, relies on assumptions embedded in the Bern multi-compartment framework itself. In strictly linear systems with parallel sinks, Stallinga (2023) [118] demonstrates mathematically that adjustment time cannot exceed residence time. The distinction exists only if one presupposes cascaded sequential reservoirs with slow intermediate equilibration due to ocean surface saturation, precisely the architecture whose empirical support is under question. In the strictly linear framework favored by the empirical fits, this is not a modeling choice but a theorem (Stallinga 2023 [118]): the burden of empirical support therefore rests with the cascaded-reservoir architecture, not with the equivalence of residence and adjustment times.

This presents a logical difficulty. When critics cite short residence times as evidence against long-term anthropogenic accumulation, the standard rebuttal invokes the residence-adjustment distinction. But that distinction presupposes the validity of the Bern multi-compartment architecture that the short residence time evidence calls into question. The argument becomes circular: the Bern model's structure is defended by a conceptual distinction that exists only within frameworks assuming that structure.

Resolving this circularity requires independent empirical tests. The isotopic evidence provides one such test. If anthropogenic emissions with $\delta^{13}C \approx -28‰$ were accumulating over decades to centuries as the Bern multi-compartment models predict, the net input isotopic signature $\delta^{13}C_I$ should shift progressively toward more depleted values as the anthropogenic fraction grows. Observed constancy of the input signal $\delta^{13}C_I \approx -13.2‰$ from the Little Ice Age through the present, despite multifold cumulative emission increases, constitutes such an independent test, one the anthropogenic accumulation hypothesis fails.

9.3 From Bern to CMIP6: Trading One Invalid Foundation for Another

The evolution of IPCC carbon cycle assessment reveals a progressive retreat from transparency without resolution of underlying physical problems. Through AR5, the Bern impulse response function provided explicit, if flawed, parameterization. The non-decaying a_0 term, the cascaded timescales, and the source-discriminating sink assumptions were at least visible for scrutiny. Critics could identify precisely where the model violated mass conservation and molecular indistinguishability.

AR6's shift to CMIP6 Earth System Model ensembles obscures these problems without solving them. The chapter justifies this transition as advancement, citing "improved representation of key

processes" including nutrient limitations, permafrost dynamics, and carbon-climate feedbacks [62]. No acknowledgment appears of the Bern model's physical flaws or the superior empirical performance of linear alternatives. The framing presents complexity as progress.

Yet the CMIP6 ESMs retain qualitatively identical long-term persistence assumptions, 20–30% of emissions remaining airborne over millennia, inherited from the same GCM/EMIC frameworks that generated the original Bern parameterizations. The cascaded reservoir architecture, the implicit source discrimination, and the exclusion of ultimate geological sinks persist within the ESM structures; they are merely buried within millions of lines of code rather than expressed in four-term analytical equations. Opacity has replaced transparency while the physics remains unchanged.

More fundamentally, the shift to ESM ensembles introduces a new and arguably more severe invalidity. All CMIP models, whether characterized as AOGCMs or ESMs, are explicitly tuned to reproduce historical GMST trajectories. This tuning target pervades model development: parameters governing cloud feedbacks, aerosol forcing, ocean mixing, and indeed carbon cycle response are adjusted until simulated GMST tracks observations within acceptable bounds. The adequacy of an ESM is judged primarily by its GMST fidelity.

This practice would be unproblematic if GMST constituted a physically meaningful quantity. It does not. Temperature is defined as $T = (\partial U/\partial S)_{V,N}$: an intensive thermodynamic property existing only in local equilibrium subsystems [37]. Intensive properties cannot be meaningfully summed or averaged across heterogeneous non-equilibrium systems. The Earth's surface atmosphere-ocean domain encompasses air-seawater density ratios exceeding 800:1, oceanic heat capacities thousands of times greater than atmospheric values per unit volume, and pervasive non-equilibrium states maintained by solar forcing, convection, and phase transitions.

Averaging local temperatures across such a system produces statistical constructs without physical uniqueness. Different averaging methods yield divergent numerical values and potentially opposite temporal trends from identical measurements. No physical law prescribes one averaging method over another; heat flows are governed by gradients, not averages, per the second law of thermodynamics. The IPCC's GMST, an area-weighted hybrid of near-surface air temperatures over land with sea surface temperatures over ocean, compounds these violations by treating intensive quantities as extensive and appending Celsius units to what are fundamentally dimensionless indices [20].

The consequence for AR6 carbon cycle conclusions is severe. ESM-derived estimates of ocean and land carbon uptake, airborne fraction projections, and the persistence characteristics that replaced explicit Bern coefficients all emerge from models tuned to a quantity that lacks physical meaning. When a model's fundamental calibration target is thermodynamically meaningless, its outputs cannot possess physical meaning regardless of their internal consistency or ensemble agreement. The apparent precision of AR6's carbon budget figures (e.g., 2.5 ± 0.6 GtC yr^{-1} ocean sink) inherits this foundational invalidity.

9.4 Thermodynamic Considerations and Their Scope

The thermodynamic refutation of GMST raises foundational questions that extend beyond carbon cycle attribution. If no physically meaningful global temperature exists, then climate sensitivity, defined as equilibrium temperature response to CO_2 doubling, becomes undefined as a physical quantity. Model-derived sensitivity estimates would represent relationships between physically meaningless statistical constructs rather than physical parameters.

This conclusion may appear too strong. Critics might argue that even if GMST lacks rigorous thermodynamic meaning, it serves as a "useful index" correlating with physical changes. However, Essex et al. (2007) [37] demonstrate that different averaging methods applied to identical temperature fields can and do yield opposite trends. An index whose trend direction depends on arbitrary methodological choices cannot reliably indicate the direction of physical change, let alone its magnitude. Furthermore, to claim that an index represents energy or energy change requires a mathematical derivation, typically a Taylor expansion or limiting argument, showing how the index captures the leading-order behavior of the physical quantity within bounded error.

The practical implications are significant. Energy balance arguments central to climate science relate radiative forcing to temperature change through sensitivity parameters. If the "temperature change" term lacks physical meaning, the entire energy balance framework requires reconstruction on different foundations, perhaps emphasizing local and regional gradients, ocean heat content, or other extensive quantities rather than intensive temperature averages (see Cohler et al., 2026 [21]).

9.5 Why Simple Models Outperform

A recurring pattern across multiple analyses is the superior empirical performance of simple, analytically tractable models compared to complex frameworks with many adjustable parameters. This pattern invites explanation.

One possibility is that climate system dynamics on policy-relevant timescales are indeed governed by relatively simple mass-balance and thermodynamic constraints, with apparent complexity arising from natural variability rather than intricate deterministic mechanisms. Under this interpretation, elaborate models introduce spurious structure while simple models correctly capture dominant physics.

An alternative interpretation holds that complex models are tuned to match multiple constraints simultaneously, creating tension that degrades performance on any single diagnostic, whereas simple models optimized for one target naturally achieve better fits on that measure. This interpretation would not favor simple models as more physically accurate, only as better optimized for particular metrics.

Distinguishing these interpretations requires examining whether simple model success is consistent across independent diagnostics. The convergence documented here, with short residence times, local temperature precedence, stable isotopic signatures, and absence of anthropogenic fingerprints emerging from separate analyses using different data and methods, suggests the former interpretation. Multiple independent diagnostics consistently favor the same physical picture, whereas coincidental optimization success seems improbable across such diverse tests.

9.6 Implications for Future Research

Based on our analysis, we conclude that several research directions warrant priority. First, the isotopic analysis should be extended to additional proxy archives and measurement sites to test the robustness of the stable net input signature finding. Second, the stochastic causality framework should be applied to additional paleoclimate reconstructions as they become available, particularly those with improved temporal resolution. Third, the thermodynamic analysis of temperature averaging should be extended to examine whether any global metric can be constructed with physical meaning, or whether climate assessment must fundamentally shift toward regional and gradient-based approaches.

More broadly, the demonstrated inadequacy of complex models tuned to thermodynamically invalid targets suggests that model development priorities may require reassessment. Increasing resolution and adding process complexity may be counterproductive if foundational metrics lack physical meaning. Resources might be better directed toward developing physically grounded alternatives to GMST and testing whether simpler models constrained by conservation laws can match or exceed complex model performance across multiple independent diagnostics.

9.7 Scope and Assumptions

The paper takes four results as inputs whose full derivations are published elsewhere: the stationarity of the Keeling net-input signature at −13.2‰ (Koutsoyiannis 2024 [78,80]); the single-exponential bomb-^{14}C decay with 17.2-yr e-folding and no permanent component (Koutsoyiannis 2024 [77], Appendix C, with the single exponential fitted on a 20-year subset of 1965–1985 of the Hua et al. 2022 series [54] and shown to extrapolate across the full 54-year record without refitting); the physical meaninglessness of GMST (Essex et al. 2007 [37]; Cohler 2025 [20]); and the Argo-based EEI physical invalidity and uncertainty exceeding ±1 W m^{-2} at 95% CI (Cohler et al. 2026 [21]). This paper operates in a local framework throughout and does not claim or rely on any global-aggregate temperature. The 2.6 W m^{-2} TOA flux imbalance derived in Section 8

sits inside the paper's own quoted 4.3 ± 3.5 W m^{-2} (1σ) raw CERES envelope [88,90]. The paper's testable predictions are enumerated in the following section with explicit falsification criteria.

10. Testable Predictions

The framework presented here yields several clear, falsifiable predictions that can be tested with ongoing observations:

- **The Keeling net-input signature** will remain stable at −13.2 ± 0.2 ‰ through at least 2035, with no detectable drift toward fossil-fuel values despite continued anthropogenic emissions. Any statistically significant departure would falsify the claim of negligible anthropogenic net contribution.
- **Absolute uptake by ocean and land** will continue to increase proportionally with atmospheric CO_2 concentration, with no evidence of saturation or slowdown. A sustained decline in absolute uptake would contradict the linear-reservoir and temperature-driven outgassing model.
- **The atmospheric CO_2 growth rate** will show no detectable deviation beyond natural interannual variability (±0.5 ppm yr^{-1}) once TOA-imbalance-driven fluxes are accounted for, including under sharp changes in anthropogenic emissions (e.g., the 6 to 7% global decline during 2020, which produced no corresponding decline in Mauna Loa growth rate). A rapid, sustained decline in atmospheric CO_2 growth rate that tracks a corresponding decline in anthropogenic emissions would falsify the nature-dominated, short-residence-time prediction.
- **Local and regional temperature changes** will continue to precede changes in atmospheric CO_2 concentration on all observable timescales (monthly to decadal), as documented by OCO-2, FLUXNET, and expanded proxy networks. Any period in which CO_2 systematically leads temperature would falsify the unidirectional causality established here.
- **Changes in CO_2 growth rate** should only occur when the constrained TOA imbalance of 2.6 W m^{-2} changes substantially. A sustained change in CO_2 growth rate not accompanied by a corresponding change in TOA imbalance would falsify the energy-balance derivation in Section 8.

These predictions are derived directly from mass conservation, the linear RRR model (>99.7% variance), the stable Keeling intercept, and the physics-constrained 2.6 W m^{-2} natural imbalance. They require no tuning and can be rigorously tested against raw instrumental and satellite data in the coming years.

11. Conclusions

11.1 Overview of the Natural-Dominance Framework

This synthesis establishes, through convergent first-principles analyses and direct observational fits achieving explained variances exceeding 95–99%, that the post-industrial rise in atmospheric CO_2 and associated climate variability are fully explicable by natural temperature-driven biogeochemical feedbacks, biospheric amplification, solar variability, and internal ocean-atmosphere oscillations, with no detectable anthropogenic contribution required (see Table 3). At the heart of this framework is a modest natural top-of-atmosphere radiative imbalance of approximately 2.6 W m^{-2}, arising from changes in planetary albedo, TSI fluctuations, post-volcanic aerosol recovery, and multidecadal ocean cycles, which quantitatively accounts for the entire observed CO_2 increase through van 't Hoff oceanic outgassing and Arrhenius-enhanced soil respiration. This provides an internally consistent physical explanation for the current state of the atmosphere, oceans, and climate system without invoking physically invalid constructs such as millennial CO_2 persistence or GMST averaging.

Table 3. Core First-Principles Falsifications of Anthropogenic CO_2 Attribution

Claim	Direct Evidence (Data + Math)	First-Principles Violation
Net CO_2 rise & $\delta^{13}C_I$ signature anthropogenic undetectable	Keeling intercept fixed at −13.2‰ for 40+ instrumental & 500 proxy years while anthropogenic emissions 3–4x since 1978	Reservoir mass balance requires anthropogenic CO_2 input < detection limit
CO_2 residence time: 3.5–4 yrs. No "adjustment time"	RRR linear (b=1) fit to raw Mauna Loa/Barrow: >99.7% variance, split-sample validation	Bern multi-compartment model tails require impossible sequential source-discriminating sinks
Temperature → CO_2 unidirectional on all time scales	Local proxies (ice cores, Phanerozoic, ENSO, OCO-2, FLUXNET): CO_2 → Temperature not observed	Assumed CO_2 → Temperature causality contradicted by timing
GMST physically meaningless	Temperature is an intensive property; infinite averaging schemes yield divergent/opposite trends on identical data	CMIP tuning to non-physical GMST invalidates all derived claims
No long-term anthropogenic persistence: Bern model is physically invalid	Linear reservoir: $\tau \approx 4$ yr e-folding time; full decay below 1 ppm in <23 yr	Non-decaying a_0 term violates mass conservation & ultimate geological sinks

11.2 Three Convergent Observational Diagnostics

Short atmospheric CO_2 residence times of approximately 3.5–4 years, derived from refined reservoir routing applied to Mauna Loa and Barrow instrumental records [77], suggest rapid cycling of any perturbation with human emissions representing only approximately 4% of annual turnover. Local temperature precedence over CO_2 variations emerges consistently from data analysis across all examined timescales from months to millions of years, with characteristic lags scaling proportionally as expected for systems with long-range dependence [79,81]. Stable net $\delta^{13}C_I$ input signatures averaging −13.2‰ persist throughout instrumental records and proxy reconstructions extending to the Little Ice Age period, demonstrating overwhelming biosphere dominance with no detectable anthropogenic fingerprint despite substantial emission growth [78]. These diagnostics are incompatible with hypotheses positing anthropogenic emissions as the primary driver of observed changes.

11.3 Falsification of the Bern Framework

Claims of prolonged anthropogenic CO_2 influence depend critically upon Bern multi-compartment modeling artifacts lacking physical foundation. The Bern impulse response function's evolving non-physical permanent fraction and unstable parameters across IPCC assessment reports, with a_0 ranging from 0.1369 in AR2 to 0.2173 in AR5/AR6 and key timescales shifting by −85% to +263% between successive reports, reflect ad hoc tuning to sustain predetermined outcomes rather than convergence toward physical measurements [68–70]. These constructs violate mass conservation principles for chemically indistinguishable molecules and require non-physical source-discriminating sink behavior. The transition to CMIP6 Earth System Model ensembles in AR6 perpetuates equivalent qualitative assumptions through increased opacity without resolving foundational limitations [62]. The single-exponential character of the bomb-^{14}C decay record, with e-folding time τ = 18.389 yr (R^2 = 99.825%) fitted to the full 1965–2019 global Hua et al. (2022) record and consistent with the $\tau \approx 17.2$ yr upper bound Koutsoyiannis (2024) [77] obtained from the 1965–1985 subset, exhibits no long tails and no permanent fraction [23]. This constitutes direct empirical falsification of the Bern models employed in IPCC Assessment Reports and their associated claims of multi-millennial anthropogenic CO_2 persistence. By requiring a non-decaying airborne fraction and multiple widely differing e-folding time scales (~41–165 years) all of which dramatically diverge from observed data, the IPCC framework is shown to be mathematically, physically, and visually incompatible with the high-fidelity $\Delta^{14}C$ observations.

11.4 Thermodynamic Invalidity of GMST and Its Downstream Consequences

The thermodynamic invalidity of GMST extends incompatibility to all global temperature-based assertions central to climate policy. As an intensive property defined only in local equilibrium, temperature cannot be meaningfully averaged across Earth's heterogeneous, non-equilibrium surface atmosphere-ocean system [20,37]. Infinite families of mathematically valid averaging methods yield divergent values and opposing trends from identical measurements, with no physical law dictating one method over another. This mathematical indeterminacy renders quantitative claims of 1.1°C anthropogenic warming, human attribution percentages, climate sensitivity estimates, future projections of 1.5–4°C warming, carbon budgets, extreme event attribution, and sea-level forecasts physically meaningless rather than merely uncertain. All major climate models, explicitly tuned to reproduce historical GMST trajectories, inherit this invalidity throughout their coupled fluid dynamics, propagating thermodynamic impossibility across all projection variables.

11.5 Why Simple Models Outperform

These results highlight the superior alignment of simple, analytically solvable models grounded in conservation laws and basic kinetics with observational data, compared to complex frameworks reliant upon unverifiable assumptions. Linear reservoir approaches achieve explained variances exceeding 99.7% for CO_2 concentration dynamics without requiring cascaded compartments, source discrimination, nonlinear saturation, or indefinite persistence terms. The consistency of such models across instrumental and proxy records underscores that elaboration diverges from rather than converges toward observational fidelity.

11.6 Airborne Fraction: Logical vs. Computational

The distinction between the physically consistent logical airborne fraction ($ABF_L \approx 6\%$), rigorously computed via convolution integrals of cumulative emissions with origin-independent short residence times from RRR) and the incompatible, presupposition-laden computational airborne fraction ($ABF_C \approx 44\%$, despite contradicting the IPCC glossary) converges with stable net isotopic signatures; unidirectional temperature → CO_2 causality across timescales; single-exponential bomb-^{14}C decay falsifying multi-millennial persistence; and thermodynamic invalidations of GMST averaging to demonstrate that natural biogeochemical feedbacks provide a complete, empirically robust explanation for observed CO_2 variations with negligible anthropogenic contribution.

11.7 The Three Independent Empirical Falsifications

The three independent empirical falsifications documented here, namely the single-exponential bomb-^{14}C decay, the stationary Keeling net-input signature across instrumental and sclerosponge records, and the three-way isotopic convergence at approximately −13‰ across instrumental, sclerosponge, and Law Dome records (Köhler et al. 2006 [75]), each constitute a direct test of anthropogenic persistence that the IPCC framework fails on the data. Accurate understanding of Earth's climate system depends upon such scientifically grounded reassessment.

11.8 Implications and Call for Reassessment

This synthesis calls for fundamental reevaluation of climate research priorities. Emphasis should shift toward transparent, empirically consistent approaches that prioritize natural variability and first-principles physics over elaborate, data-divergent constructs and physically invalid models. The demonstrated efficacy of simple models in achieving superior data alignment, combined with the thermodynamic impossibility of current core metrics and the empirical falsification of anthropogenic carbon cycle dominance, indicates that prevailing assessment frameworks require reconstruction from foundations rather than incremental refinement.

Appendix A. NASA CERES Uncertainty-Budget Citation Chain Audit

Appendix A.1. Scope

Appendix A audits the uncertainty-budget citation chain behind the CERES-based "observed" Earth Energy Imbalance claim propagated by the IPCC Sixth Assessment Report (AR6, Chapter 7) and the EBAF Edition 4.0 data product. Every quantitative statement in this appendix is taken from the primary source cited. Independent corroboration of the broader IPCC EEI "fraud" pattern is provided by Clauser (2026) [19].

Appendix A.2. Physics and Channel Definitions

CERES measures three filtered broadband radiances [88], p. 896: shortwave (SW, 0.3 to 5 μm, reflected solar); total (TOT, 0.3 to 200 μm, reflected solar plus emitted thermal); and window (WN, 8 to 12 μm, a narrow atmospheric-window diagnostic covering only a slice of the longwave spectrum). The WN channel is not used to derive broadband LW for the net TOA budget. There is no dedicated broadband LW channel. Broadband LW radiance is instead derived by subtraction [88], p. 897: *"Since there is no LW channel on CERES, LW daytime radiances are determined from the difference between the TOT and SW channel radiances."* Nighttime LW comes from the TOT channel alone (no solar component). Priestley et al. (2002) [101], p. 2374 states the direct consequence: *"Daytime LW measurements are determined by subtracting the SW channel measurements from the TOT channel measurements, thus any inconsistencies in the calibration of the SW and SW portion of the TOT channel, SW/TOT, result in daytime LW errors."* SW and derived-daytime-LW uncertainties are therefore not independent, and any SW calibration shift enters the net TOA budget twice: once as SW itself and once in the derived LW.

Appendix A.3. Pre-Launch Requirements and the GUM Convention

The CERES pre-launch accuracy goals, stated in Wielicki et al. (1996) [130] and Wielicki et al. (1995) [131], are 1% for the shortwave reflected radiance and 0.5% for the earth-emitted radiance. Both figures are stated as accuracy requirements with no confidence-interval qualifier. Under the standard metrology convention that an unqualified uncertainty figure is a standard uncertainty, that is, 1σ (codified by the Comité International des Poids et Mesures in Recommendation INC-1 (1980), first published as ISO Guide to the Expression of Uncertainty in Measurement (GUM) 1993, adopted as US national practice in NIST Technical Note 1297 (1994), and in current authoritative form as JCGM 100:2008 [63]), the Wielicki goals are 1% SW and 0.5% LW at 1σ. The convention was in international force before both Wielicki papers were published. Converting to a 95% coverage interval ($k = 2$) gives 2% SW and 1% LW at 95% CI. These two statements are mathematically identical. They refer to the same budget.

Appendix A.4. Priestley 2002 Verbatim: The Goals Were Met and Beaten

Priestley et al. (2002) [101] reports early on-orbit validation of the *Terra* FM-1 and FM-2 instruments through September 2000. The paper's conclusion:

> *This collection of validation activities has rigorously demonstrated that pre-launch calibration goals of 1-percent for earth reflected solar and 0.5-percent for earth emitted radiance measurements have been satisfied in the Terra Edition 1 instrument and ERBE-Like data products.*
>
> — Priestley et al. (2002) [101], p. 2376

The specific achieved values are reported in Tables A1 and A2:

Table A1. Ground-to-flight absolute traceability shifts for CERES *Terra* FM-1 and FM-2 instruments, defined as "the amount of drift seen since launch." Values from Priestley et al. (2002) [101], p. 2373, compared with Wielicki et al. (1996) [130] pre-launch goals.

Channel	FM-1 (%)	FM-2 (%)	Wielicki goal (%)
TOT (LW)	0.20	0.12	0.5
SW	0.33	0.01	1.0

Table A2. Post-launch stability over the first seven months of science data collection for CERES *Terra* FM-1 and FM-2 instruments. Values from Priestley et al. (2002) [101], p. 2373.

Channel	FM-1 (%)	FM-2 (%)
TOT (LW)	0.18	0.25
SW	0.001	0.06

Unfiltering errors (page 2373): less than 0.5% SW and less than 0.4% LW.

***Terra*/TRMM inter-satellite agreement** (page 2374):

• SW 0.4% difference with 95% CI of ±0.4%, "not statistically significant";

• LW 0.4 to 0.5% day and 0.1% night with 95% CI of ±0.1%.

FM-1/FM-2 same-platform agreement (page 2374): less than 0.5% in SW and LW.

Every per-channel measured value in [101] is *under* the Wielicki et al. (1996) [130] pre-launch goal. The combined RSS at 1σ using [101]'s own numbers:

• SW: $\text{sqrt}(0.33^2 + 0.001^2 + 0.5^2 + 0.5^2) = 0.75\%$ at $1\sigma = 0.75$ W m^{-2} on 100 W m^{-2} SW

• LW: $\text{sqrt}(0.20^2 + 0.18^2 + 0.4^2 + 0.5^2) = 0.70\%$ at $1\sigma = 1.68$ W m^{-2} on 240 W m^{-2} LW

• Net RSS at $1\sigma \approx 1.85$ W m^{-2}

Or using only absolute traceability (strictest reading):

• SW: 0.33% at $1\sigma = 0.33$ W m^{-2}

• LW: 0.20% at $1\sigma = 0.48$ W m^{-2}

• Net RSS at $1\sigma \approx 0.58$ W m^{-2}

[101] on-orbit validation therefore gives a defensible per-instrument absolute uncertainty of 0.58–1.85 W m^{-2} at 1σ, depending on which components are included. Both values are *tighter* than the pre-launch goals.

Appendix A.5. Priestley 2011 Verbatim: Seven-Year Flight Record

Priestley et al. (2011) [102] reports the *Terra* FM-1 and FM-2 calibration record through April 2007. The paper documents three structural facts that bear on absolute accuracy.

First, the Edition 2 product is characterized (abstract, page 3) as *"available for science investigations for which an accuracy of 2% is sufficient."* This is a floor-of-usability statement, not a delivered-accuracy figure. The paper does not provide a per-channel delivered absolute accuracy.

Second, FM-1 and FM-2 fly on the same spacecraft and view each nadir pixel within 3.3 s of each other. After all Edition 2 corrections, the paper reports persistent same-platform residuals (page 16): *"For edition 2 the difference of nighttime fluxes of FM-1 and FM-2 are steady at 0.7 W m^{-2} after 2003 and the daytime longwave fluxes agree across the data period to a fraction of a W m^{-2}. However, the shortwave fluxes disagree by up to 2 W m^{-2}."* The RSS of these persistent residuals is $\text{sqrt}(2^2 + 1^2) \approx 2.24$ W m^{-2} at 1σ broadband. This is a direct observation of per-instrument error between two instruments viewing the same scenes 3.3 seconds apart, and it is a floor on true per-instrument uncertainty regardless of what any stated calibration budget claims.

Third, these residuals cannot be removed. The paper states (page 16): *"A change of shortwave channel gain of either FM-1 or -2 to bring them into agreement will create a disagreement in the daytime longwave flux. Likewise, a change of total channel gain will create disagreement in nighttime longwave fluxes."* The residuals are structural, not random noise.

The paper also notes (page 19) that Edition 2 does not correct for *"changes of the spectral response of the longwave portion of the total channels and the spectral responses of the shortwave channels."* These uncorrected spectral response changes propagate into both SW and derived LW.

Appendix A.6. Loeb 2009 Cites Priestley 2002 Incorrectly

Loeb et al. (2009) [91] introduces the "objective constrainment algorithm" that shifts CERES SW and LW TOA fluxes to *force* agreement with the Argo-based OHC modeled energy balance. Section 4 of the paper, in setting up its uncertainty budget, states:

> *The CERES absolute calibration uncertainty is ±2% in the SW channel and ±1% in the TOT channel, at the 95% confidence level (Priestley et al. 2002).*
>
> — Loeb et al. (2009) [91], Section 4

Priestley et al. (2002) [101] contains no such statement. The paper does not use the phrase "2% at 95% confidence level" anywhere in its text. The specific 95% confidence intervals that [101] does report are the ±0.4% CI on the *Terra*/TRMM SW comparison and the ±0.1% CI on the *Terra*/TRMM LW comparison, both for inter-satellite comparisons rather than absolute calibration.

The numbers [91] attributes to [101] are mathematically identical to the pre-launch goals multiplied by two (converting 1σ to 2σ under the GUM convention). Loeb et al. (2009) [91] therefore cites the pre-launch *goals* under the name of Priestley et al. (2002) [101], but uses them as the "*actual uncertainty*" of the delivered instrument. However, [101]'s measured values are three to six times tighter than the values [91] attributes to it.

Appendix A.7. Loeb 2009 Constrainment Arithmetic and the Hansen Admission

[91] reports the raw SRBAVG-GEO Ed2D net TOA flux as 6.5 W m^{-2}. It forces this value to match the modeled Earth energy imbalance target of 0.85 W m^{-2} from Hansen et al. (2005) [49]. The required adjustment is 5.65 W m^{-2}.

Hansen et al. (2011) [48] admits the forcing explicitly (as compiled by Clauser (2026) [19]):

> *The precision achieved by the most advanced generation of radiation budget satellites is indicated by the planetary energy imbalance measured by the ongoing CERES (Clouds and the Earth's Radiant Energy System) instrument (Loeb et al., 2009), which finds a measured 5-yr-mean imbalance of 6.5 W m^{-2} (Loeb et al., 2009). Because this result is implausible, instrumentation calibration factors were introduced to reduce the imbalance to the imbalance suggested by climate models, 0.85 W m^{-2} (Loeb et al., 2009).*
>
> — Hansen et al. 2011 [48], p. 13443

This is an on-the-record admission that the CERES satellite data were adjusted to match a *modeled* target, not that a model was adjusted to match satellite data. The direction of causation runs from model to measurement, which inverts standard scientific procedure.

Appendix A.8. Raw Value, Constrained Value, and Target Decomposition

Loeb et al. (2018) [88] produces the EBAF Edition 4.0 TOA product. Table A3 reproduces Table 5 of that paper, listing the global-mean TOA fluxes for July 2005 through June 2015 before and after application of the constrainment algorithm.

Table A3. Global-mean TOA fluxes (W m^{-2}) from CERES EBAF Ed4.0 before and after the "objective constrainment" adjustment, July 2005–June 2015. Reproduced from Loeb et al. (2018) [88], Table 5. We compute the shift column as (adjusted − unadjusted).

Channel	Unadjusted (W m^{-2})	Adjusted (W m^{-2})	Shift (W m^{-2})
SW outgoing	97.1	99.1	+2.0
LW outgoing	238.6	240.1	+1.5
Net imbalance	4.3	0.71	−3.59

The adjusted net value of 0.71 W m^{-2} is reported with ±0.10 W m^{-2} at 95% confidence. [88], p. 904 states the provenance of this value:

> *In the current version, the global annual mean values are adjusted such that the July 2005–June 2015 mean net TOA flux is 0.71 ± 0.10 W m^{-2}, as provided in Johnson et al.*

> *(2016) [66] [uncertainties at the 95% confidence level account for expendable bathythermographs (XBT) correction uncertainties and Argo sampling errors for 0–1800 m]. The uptake of heat by Earth for this period is estimated from the sum of (i) 0.61 ± 0.09 W m⁻² from the slope of weighted linear least squares fit to Argo OHCA data to a depth of 1800 m analyzed following Lyman and Johnson (2008), (ii) 0.07 ± 0.04 W m⁻² from ocean heat storage at depths below 2000 m using data from 1981–2010, and (iii) 0.03 ± 0.01 W m⁻² from ice warming and melt and atmospheric and lithospheric warming from 1971–2010.* — Loeb et al. (2018) [88], p. 904

The sum 0.61 + 0.07 + 0.03 = 0.71 W m⁻², the claimed "observed" EEI value, is therefore not a CERES satellite measurement. It is a sum of ocean and cryosphere purported heat uptake estimates derived from Argo and other in situ instruments, substituted for the CERES data. The CERES data are then simply replaced.

Loeb 2018 [88] denies this. The same page (p. 904) states: *"we use the objective constrainment algorithm described in Loeb et al. (2009) to adjust SW and LW TOA fluxes within their ranges of uncertainty to remove the inconsistency between average global net TOA flux and heat storage in the earth-atmosphere system."*

The word "objective" in Loeb 2009 [91]'s "objective constrainment algorithm" has a specific mathematical meaning. It is a Lagrangian least-squares minimization that, given the measured net flux, the target net flux, and the per-channel uncertainties σ_{SW} and σ_{LW}, produces unique closed-form shifts ΔF_{SW} and ΔF_{LW} that sum exactly to the required net correction and whose ratio equals $\sigma_{SW}^2/\sigma_{LW}^2$ exactly.

Table 5 of [88] itself proves the algorithm was *never* applied. The displayed shifts of +2.0 W m⁻² SW and +1.5 W m⁻² LW sum to 3.5, which is 0.09 short of the 3.59 W m⁻² required to move the net TOA imbalance from 4.3 to 0.71, and their 4:3 ratio contradicts what the Loeb 2009 [91] Lagrangian would output from the stated per-channel calibration uncertainties. At the stated σ_{SW} = 1.0 W m⁻² and σ_{LW} = 1.8 W m⁻², the Lagrangian would produce ΔF_{SW} = 0.85 and ΔF_{LW} = 2.74 W m⁻², not the published 2.0 and 1.5.

[88]'s two claims are therefore directly contradictory. An objective algorithm's output is determined by its inputs. It cannot be arranged to equal a target "provided" externally.

The Argo-derived target itself fails two independent falsification tests, documented in detail in Cohler et al. (2026) [21]. First, categorical physical invalidity: the Argo construction treats OHC-anomaly as a globally additive quantity derived from differences of an intensive variable (temperature) against arbitrary climatological baselines, and Argo samples only the upper ~2000 m while excluding roughly half of ocean volume and thermal energy, horizontal positions are recorded only at surfacing with offsets up to 50 km and 10 hours, and CERES itself is adjusted to match Argo therefore provides no independent validation. Second, uncertainty analysis conducted arguendo while accepting the non-physical construction: combining mesoscale variability (±0.9 W m^{-2}), deep-ocean undersampling (±0.35 W m^{-2}), sea-level-budget closure discrepancy (±0.33 W m^{-2}), Eulerian–Lagrangian discrepancy (±0.25 W m^{-2}), baseline arbitrariness (±0.2 W m^{-2}), and polar undersampling (±0.1 W m^{-2}) yields a total uncertainty of greater than ±1 W m^{-2} at 95% CI, approximately an order of magnitude larger than the ±0.10 W m^{-2} figure claimed by Johnson et al. (2016) [66] and Loeb et al. (2018) [88] and finally adopted by AR6. Under either reading, the Argo-derived target that CERES is forced to match is itself statistically indistinguishable from zero, which is further corroborated by the independent analysis of Clauser (2026) [19].

Appendix A.9. Uncertainty Figures: Global Versus Regional

[88] does not state a single comprehensive global 1σ uncertainty on the net TOA flux. The paper provides two separable figures that bear on the detection question.

Global calibration-only 1σ (Introduction, p. 897): *"The actual uncertainty for CERES resulting from calibration alone is 1% SW and 0.75% LW radiation [one standard deviation (1σ)], which corresponds to 2 W m⁻², or 0.6% of the total TOA outgoing radiation."* Combining 1% × 100 W m^{-2} SW and 0.75% × 240 W m^{-2} LW in RSS gives sqrt($1^2 + 1.8^2$) = 2.06 W m^{-2} at 1σ. This figure covers calibration only. It excludes radiance-to-flux conversion error, time

interpolation error, and spectral response changes. The true total global 1σ is therefore larger than 2 W m^{-2}, but is not stated as a single number anywhere in the paper.

Regional 1°×1° monthly 1σ (Table 8, p. 915). The Table 8 figures (all-sky net 4.2 W m^{-2} *Terra*-only, 3.5 W m^{-2} *Terra–Aqua*) are regional monthly uncertainties on individual 1°×1° grid cells, not global uncertainties. Section 4 builds these totals from component uncertainties and cites Su et al. (2015) [123] correctly for the angular distribution model radiance-to-flux conversion component. The calibration component (1 W m^{-2} for SW and 1.8 W m^{-2} for LW) is stated without citation in Sections 4a and 4b.

The EBAF uncertainty bar compared to CERES. The EBAF value of 0.71 ± 0.10 W m^{-2} at 95% CI (equivalent to ±0.05 W m^{-2} at 1σ) is global and annual. The appropriate comparison is to the global calibration-only 1σ of 2 W m^{-2}, which gives a factor of 40 by which the EBAF bar is tighter than the CERES calibration budget alone. The tightness of the EBAF uncertainty comes from the Argo and the in situ purported heat uptake estimates, not from CERES.

Appendix A.10. Inferring the Actual Global 1σ From Loeb 2018's Own Numbers

Although [88] does not provide any total global 1σ net TOA uncertainty, two independent inference methods using the paper's own quantitative claims give consistent estimates. Both methods start from quantities [88] does state and derive the global 1σ from them.

Method 1. Lower bound from the "constrainment within uncertainty" claim.

[88], p. 904 describes the adjustment procedure: *"we use the objective constrainment algorithm described in Loeb et al. (2009) to adjust SW and LW TOA fluxes within their ranges of uncertainty to remove the inconsistency between average global net TOA flux and heat storage in the earth-atmosphere system, as determined primarily from ocean heat content anomaly (OHCA) data."*

The two per-channel shifts shown above in Table A3 are co-directional (both increase outgoing radiation) and were chosen specifically to produce the desired net reduction. The claim that this chosen combination is "within their ranges of uncertainty" sets a direct lower bound on the global net 1σ uncertainty.

Why the net floor is a simple sum of 2.0 + 1.0 = 3.5 W m^{-2}, not the RSS of 2.5 W m^{-2}.

Two independent physical facts force the linear-sum combination rather than independent-error quadrature:

Fact 1. SW and LW uncertainties are not independent because LW is derived from TOT minus SW. From Appendix A.2, the daytime LW radiance is calculated as LW = TOT − SW, so any error in the SW channel measurement propagates into the derived LW with the opposite sign. Writing the net TOA flux explicitly:

• Daytime pixel:

Net = Incoming − SW − LW = Incoming − SW − (TOT − SW) = Incoming − TOT

• Nighttime pixel:

Net = 0 − 0 − TOT = −TOT

At the raw measurement level, the SW channel calibration error cancels exactly out of the net because SW appears with opposite signs in the SW flux (+) and in the derived LW flux (−). Treating SW and LW flux uncertainties as independent and combining them in quadrature therefore overstates the net uncertainty from the calibration component. The physically correct combination for correlated error sources with same-sign effect on the net is linear sum, not RSS.

Fact 2. The chosen shifts are co-directional, not random realizations. The constrainment procedure chose SW = +2.0 and LW = +1.5 specifically because together they move net by the required 3.5 W m^{-2} (but [88] actually just substituted 3.59 W m^{-2} and ignored its own claimed "objective" constrainment). These are not independent draws from SW and LW error distributions. They are a single coordinated adjustment whose net effect on the budget quantity is the linear sum. For a coordinated chosen combination to be "within net 1σ", the net 1σ must be at least as large as the combined net effect.

Both facts converge on the same answer: $\sigma_{\mathrm{Net}_{\mathrm{Global}}} \geq 3.5$ W m^{-2}.

Per-channel constraints additionally reveal that the stated SW 1σ is understated.

Taking the per-channel reading of "within range of uncertainty" at the standard metrology default of 1σ:

• $\sigma_{\mathrm{SW}_{\mathrm{global}}} \geq 2.0$ W m^{-2} (2% of 100 W m^{-2} SW baseline)

• $\sigma_{\mathrm{LW}_{\mathrm{global}}} \geq 1.5$ W m^{-2} (0.625% of 240 W m^{-2} LW baseline)

The implied minimum global SW 1σ of 2.0 W m^{-2} is twice the [88] Introduction's stated figure of 1 W m^{-2} (= 1% × 100 W m^{-2}). One of two things must be true: (a) the stated 1% SW figure is understated and the actual SW 1σ is at least 2%, or (b) the SW shift was not actually "within 1σ" and the paper's own characterization of the adjustment fails. Either reading destroys [88]'s internal consistency.

The implied LW 1σ of 1.5 W m^{-2} is close to the stated 1.8 W m^{-2}, consistent with the LW budget having been inflated to accommodate the required LW shift. The 0.75% LW figure is examined in detail in Appendix A.11.

Lower bound from the constrainment logic: global net 1σ ≥ 3.5 W m^{-2}.

Method 2. Derivation from the Table 8 regional components.

[88] Sections 4a and 4b decompose the Table 8 all-sky regional 1°×1° monthly 1σ uncertainties into three components. For the *Terra*-only period, the per-component values are given in Table A4.

Table A4. Decomposition of the Loeb et al. (2018) [88] Table 8 all-sky regional 1°×1° monthly 1σ uncertainties for the *Terra*-only period into physical components. All values in W m^{-2}, regional 1σ.

Component	SW	LW	Correlation structure
Diurnal correction / temporal interpolation	2.7	2.2	Partially systematic (cloud diurnal patterns)
Radiance-to-flux (ADM, [123])	1.0	0.75	Partially systematic (scene-type biases)
CERES calibration	1.0	1.8	Fully systematic (instrument-wide)
Regional 1σ total (RSS)	3.0	3.0	Matches Table 8

Going from regional 1°×1° monthly to global annual mean depends on the spatial and temporal correlation of each component. Three bounding cases:

Case 2A, upper bound (fully systematic). All three components are globally correlated and do not average down. Global 1σ equals regional 1σ. Net global 1σ = 4.2 W m^{-2} (*Terra*-only) or 3.5 W m^{-2} (*Terra–Aqua*).

Case 2B, realistic. Calibration is fully systematic (does not average down). Radiance-to-flux reduces by roughly 50% on global averaging (scene-type correlations partly cancel). Diurnal correction reduces by roughly 25% (cloud diurnal patterns are correlated by latitude band). Then:

• Global SW 1σ = sqrt($2.0^2 + 0.5^2 + 1.0^2$) = sqrt(5.25) = 2.29 W m^{-2}

• Global LW 1σ = sqrt($1.65^2 + 0.4^2 + 1.8^2$) = sqrt(6.12) = 2.47 W m^{-2}

• Global Net 1σ = sqrt($2.29^2 + 2.47^2$) = sqrt(11.35) = 3.37 W m^{-2}

Case 2C, lower bound (calibration only). All non-calibration components average down to zero on global annual mean. Then global 1σ reduces to calibration alone:

Global Net 1σ = sqrt($1.0^2 + 1.8^2$) = 2.06 W m^{-2}

Case 2C matches the [88] Introduction's stated calibration-only global 1σ of 2 W m^{-2}. This confirms the Introduction figure is consistent with [88], Table 8 only under the most generous assumption that all non-calibration regional uncertainties cancel perfectly on global averaging. That assumption is not physically defensible for correlated systematic effects.

Method 3. Independent floor from FM-1/FM-2 same-platform residuals.

Priestley et al. (2011) [102] documents persistent Edition 2 residuals between FM-1 and FM-2, two CERES instruments on the same *Terra* spacecraft viewing each nadir pixel within 3.3 s of each other. After all Edition 2 corrections, the residuals are 2 W m^{-2} daytime SW and 1 W m^{-2} nighttime LW. These are direct measurements of inter-instrument disagreement on the same scene, not model inferences.

If FM-1 reads $F + e_1$ and FM-2 reads $F + e_2$ for the same true flux F, then (FM-1 − FM-2) = $e_1 - e_2$. Under the IID treatment where both instruments have independent 1σ error σ, the per-instrument 1σ equals the observed difference divided by sqrt(2):

• Per-instrument $\sigma_{SW} \geq 2.0 / \text{sqrt}(2) = 1.41$ W m^{-2}

• Per-instrument $\sigma_{LW} \geq 1.0 / \text{sqrt}(2) = 0.71$ W m^{-2}

• Per-instrument $\sigma_{\text{Net RSS}} \geq \text{sqrt}(1.41^2 + 0.71^2) = 1.58$ W m^{-2}

If the residuals are persistent (same sign over years, as the paper reports), the IID assumption fails and the floor is higher. In the limit where one instrument is correct and the other carries the entire disagreement, the per-instrument error is the full residual:

• Per-instrument $\sigma_{SW} \geq 2.0$ W m^{-2}

• Per-instrument $\sigma_{LW} \geq 1.0$ W m^{-2}

• Per-instrument $\sigma_{\text{Net RSS}} \geq \text{sqrt}(2.0^2 + 1.0^2) = 2.24$ W m^{-2}

The FM-1/FM-2 residual is **not independent** of the calibration and radiance-to-flux components in Method 2, so it **cannot be added in quadrature** to those. It is a direct manifestation of the same instrumental errors from a different angle. What it establishes is that the stated 1% SW calibration budget (= 1 W m^{-2} per instrument at 1σ) is understated by at least a factor of 1.4 (IID treatment) or 2 (persistent treatment). The observation that two instruments disagree by 2 W m^{-2} while the stated per-instrument 1σ is 1 W m^{-2} is prima facie evidence that the stated budget does not describe the actual instrument performance.

Method 3 is therefore a cross-check on Methods 1 and 2, not an additional term in the quadrature.

Synthesis.

Combining all three methods, the inferred global net 1σ net TOA uncertainty from [88]'s own quantitative claims and the direct FM-1/FM-2 observation in Priestley et al. (2011) [102] is summarized in Table A5.

Table A5. Inferred global net 1σ net TOA uncertainty from [88]'s own quantitative claims and the direct FM-1/FM-2 observation in [102]. The strongest single lower bound is highlighted in bold.

Inference	Global net 1σ (W m^{-2})	Nature
Loeb 2018 Introduction, calibration only	2.00	Stated figure, lower bound on total
Method 2 Case 2C, calibration-only reduction	2.06	Consistent with Introduction, lower bound
Method 3, FM-1/FM-2 persistent treatment	2.24	Direct observation, per-instrument floor
Method 2 Case 2B, realistic reduction factors	3.37	Realistic from Table 8 decomposition
Method 1, constrainment-within-1σ claim	**3.59**	**Direct lower bound from Loeb 2018** [88]**'s own claim**
Method 2 Case 2A, fully systematic (*Terra*-only)	4.20	Upper bound; regional 1σ = global 1σ

The true global 1σ falls in the range 2 to 4.2 W m^{-2}. The strongest single result is from Method 1: [88]'s own claim that the 3.59 W m^{-2} net shift was done "within range of uncertainty" requires,

by their own logic, that $\sigma_{\text{net}_{\text{global}}} \geq 3.59$ W m^{-2}. However, [88] reports none of these numbers except the calibration-only lower bound of 2 W m^{-2} and the regional monthly figures. The absence of a stated total global 1σ does not appear to be an oversight. Reporting any value in this range would make the detection claim collapse (0.71 / 3.5 = 0.20σ above zero, well below any detection threshold) and would expose that even the most "generous" reading of the constrainment procedure cannot both make the adjustment valid and keep the observed EEI value detectable.

Appendix A.11. The 0.75% LW Calibration Figure: 50% Loosening of Wielicki Pre-Launch Goal

[88] states, with no citation attached:

> *The actual uncertainty for CERES resulting from calibration alone is 1% SW and 0.75% LW radiation [one standard deviation (1σ)], which corresponds to 2 W m⁻², or 0.6% of the total TOA outgoing radiation.*
>
> — Loeb et al. (2018) [88], p. 897

This 0.75% LW at 1σ is 50% looser than Wielicki pre-launch goal of 0.5% LW at 1σ, and 3.75 times looser than [101]'s achieved absolute LW traceability of 0.20% at 1σ. No reference is attached to justify this loosening. The number appears nowhere in the prior CERES calibration literature. Comparison is given in Table A6.

Table A6. Comparison of stated CERES calibration uncertainty figures (SW and LW channels, 1σ) across the primary literature. The 0.75% LW value appearing in [88] is 50% looser than the Wielicki 1996 pre-launch goal and 3.75 times looser than [101]'s documented FM-1 absolute LW traceability. Wielicki and Loeb values are 1σ instrument accuracy specifications. Priestley values are measured ground-to-orbit calibration shifts (|Δ|, with 95% CI bounds ≤ 0.04% in all cases, hence 1σ on the shifts ≤ 0.02%). Both represent achieved or targeted absolute accuracy.

Source	SW accuracy (%)	LW accuracy (%)	Status
Wielicki 1996 [130] pre-launch goal	1.00	0.50	Published
Priestley 2002 [101] achieved, FM-1 traceability	0.33	0.20	Published, under goal
Priestley 2002 [101] achieved, FM-2 traceability	0.01	0.12	Published, under goal
Loeb 2009 [91] (attributed to Priestley 2002)	1.00	0.50	Matches Wielicki goals
Loeb 2018 [88]	1.00	0.75	LW silently loosened, uncited

The value 0.75% is not derived from any instrument validation. It may not be coincidence that the LW channel adjustment falls just inside 1σ. The required LW adjustment is 1.5 W m^{-2} (Table A3). For the "within range of uncertainty" claim to survive the LW arithmetic, the 1σ LW uncertainty must satisfy $\sigma_{\text{LW}} \geq 1.5$ W m^{-2}. Expressed as a percentage of 240 W m^{-2} mean LW flux, this requires $\sigma_{\text{LW}} \geq 0.625\%$. The figure 0.75% is the smallest round value above 0.625% that keeps the LW adjustment inside 1σ. LW verification using 1.5 W m^{-2} shift:

• [130] 0.5% ⇒ $\sigma_{\text{LW}} = 1.20$ W m^{-2}; $1.5/\sigma_{\text{LW}} = 1.25$ → OUTSIDE 1σ

• [101] 0.20% ⇒ $\sigma_{\text{LW}} = 0.48$ W m^{-2}, $1.5/\sigma_{\text{LW}} = 3.13$ → DESTROYS claim

• [88] 0.75% ⇒ $\sigma_{\text{LW}} = 1.80$ W m^{-2}, $1.5/\sigma_{\text{LW}} = 0.83$ → inside 1σ, claim survives

The SW budget was not similarly loosened. 1% SW matches the Wielicki pre-launch goal, and any further loosening would be visible against prior literature. The adjustment was loaded entirely onto the derived LW channel, which has fewer external benchmarks. The LW channel is not even measured directly in the NASA CERES products; it is computed by subtracting SW from TOT, so loosening the LW budget hides inside a derived quantity.

Appendix A.12. Even After the Loosening, the Net Adjustment Remains Outside 1σ

Taking [88]'s own calibration budget at face value:

SW shift / σ_{SW} = 2.0 / (1% × 100) = 2.0 / 1.0 = 2.00σ ← OUTSIDE 1σ

LW shift / σ_{LW} = 1.5 / (0.75% × 240) = 1.5 / 1.8 = 0.83σ ← barely inside

Net (co-directed) shift = 3.5 W m^{-2}

Net calibration 1σ (RSS) = sqrt($1.0^2 + 1.8^2$) = 2.06 W m^{-2}

Net shift / net σ = 3.5 / 2.06 = 1.70σ ← OUTSIDE 1σ

Against [101] actual-achieved net 1σ of 0.58 W m^{-2}:

Net shift / net σ = 3.5 / 0.58 = 6.03σ ← complete destruction

The SW shift is already outside 1σ using [88]'s own budget. The 0.75% LW loosening is cosmetic. It makes the per-channel LW arithmetic look "valid" at channel-by-channel inspection, but the net adjustment remains at 1.70σ when the two shifts are combined in RSS, which is outside any standard "within 1σ" claim.

Appendix A.13. The σ-Count Collapse of the Detection Claim

The reported "observed" value of 0.71 W m^{-2} is a purported global and annual EEI [88]. The relevant comparison is to a global 1σ uncertainty for which several candidate values are given in Table A7.

Table A7. σ-count of the reported "observed" EEI value of 0.71 W m^{-2} against candidate global 1σ CERES uncertainties from primary sources.

Uncertainty source (global, 1σ)	1σ (W m^{-2})	0.71 / 1σ	Status
[101] achieved best, strictest reading	0.58	1.22σ	Marginal, but Adjustment is 6σ
[88] Introduction (p. 897), calibration only, lower bound	2.00	0.36σ	Undetectable
Priestley 2011 [102] FM-1/FM-2 structural floor	2.24	0.32σ	Undetectable
L'Ecuyer 2015 [85] independent assessment	5.0	0.14σ	Undetectable
Stephens and L'Ecuyer 2015 [120] joint	5.6	0.13σ	Undetectable

[88], Table 8 all-sky net regional 1°×1° monthly uncertainties (4.2 W m^{-2} *Terra*-only, 3.5 W m^{-2} *Terra–Aqua*) are not included in Table A7 because they are regional monthly and not directly comparable to a global annual claim unless one assumes case 2A explained in Appendix A.10.

In every row except the strictest reading of [101], the reported "observed" imbalance of 0.71 W m^{-2} is below 0.4σ above zero and therefore not detectable against the stated satellite uncertainty. In the row where it is marginally detectable, the 3.5 W m^{-2} adjustment that produces it is 6σ outside the uncertainty, which invalidates the method that generated the value. The two claims cannot both be true. If the uncertainty is small enough to make 0.71 detectable, the adjustment is far outside uncertainty. If the uncertainty is large enough to swallow the adjustment, 0.71 is not detectable. Had [88] reported a total global 1σ net TOA uncertainty, this contradiction would have been devastating.

Appendix A.14. Loeb 2020 Explicit Anchoring to In Situ Data

Loeb et al. (2020) [90] confirms the anchoring directly. The paper applies *"an offset to the satellite EEI such that its mean value over the 15-year period"* matches the in situ ocean mean. It states that *"satellite incoming and outgoing radiative fluxes are presently not at the level of accuracy required"* to resolve EEI absolutely. The CERES absolute value is therefore replaced by the Argo calculation.

Appendix A.15. Retraction History Ignored by IPCC AR6

Stephens et al. (2012) [121] reported "Top-of-Atmosphere EEI = 0.6 ± 0.4 W m^{-2}" based on the Lyman and Loeb analysis of ocean heat content anomaly data. This value was adopted by IPCC AR5 (2013) [59] as its central estimate.

Three years later, two of the [121] coauthors retracted the result. L'Ecuyer et al. (2015) [85] re-analyzed the data with ocean heat content constraints and reported "EEI = 0 ± 5.0 W m^{-2}", fully consistent with no imbalance. Stephens and L'Ecuyer (2015) [120] jointly issued what Clauser 2026 [19] calls the authors' confession: EEI = 0 ± 5.6 W m^{-2} for both satellite and OHCA datasets, with the explicit statements that the "adjustments" needed to produce the earlier result were *"by about 10 W m^{-2}"* relative to the claimed imbalance; that the choice of which dataset to adjust was made *"in a totally ad hoc"* fashion; and that *"there is no real evidence to support one adjustment approach over the other."*

IPCC AR6 (2021) [61] cites the L'Ecuyer 2015 retraction but adopts neither the EEI = 0 ± 5.0 value nor the Stephens and L'Ecuyer 2015 EEI = 0 ± 5.6 "confession." AR6 instead propagates 0.7 ± 0.2 W m^{-2}, a value very similar to the Loeb et al. (2018) [88] replacement value of 0.71 ± 0.10 W m^{-2}, which purportedly depends on exactly the "adjustment" procedure that L'Ecuyer and Stephens had publicly abandoned six years earlier.

Appendix A.16. Physical Mechanism of the Raw-Value Error

The 4.3 W m^{-2} raw CERES EEI value is not a valid detection of a physical energy imbalance. It is an instrumental artifact identified by Clauser (2026) [19] as a specific design flaw: the CERES line-scan imager has a fan-shaped field of view approximately ±52° East-West and only approximately ±0.2° North-South. Specular reflection from the ocean surface (sun glare), which is the dominant short-wave return at shallow solar incidence angles and is highly directional, appears as a bright narrow stripe oriented radially from the sub-solar point. The stripe is almost never aligned with the East-West scan direction, so the glare signal is systematically under-sampled. The under-reporting of outgoing SW produces an apparent positive net TOA imbalance that is instrumental, not physical. The constrainment algorithm covers this hardware flaw with a model-matched adjustment rather than correcting it.

Clauser's instrumental audit establishes that the raw CERES measurement is biased high through the sun-glare and viewing-angle flaw. [88] Argo-matched 0.71 W m^{-2} reflects a replacement by invalid OHC anomaly data, not independent radiometric measurement (see Appendix A.8, [21]). The CO_2 mass-balance derivation in Section 8 requires a natural TOA imbalance of approximately 2.6 W m^{-2} to account for the observed atmospheric CO_2 growth rate through van 't Hoff oceanic outgassing and Arrhenius soil respiration kinetics. The 2.6 W m^{-2} value lies below the raw CERES 4.3 W m^{-2}, consistent with Clauser's finding that the raw figure is biased high. It also lies above the Argo-derived EEI of 0.71 W m^{-2}. Neither endpoint is defensible on its own merits.

Appendix A.17. Summary Table of the Citation Chain

Table A8 summarizes the citation-chain failures and uncited claims documented in Appendices A.3 through A.15.

Appendix A.18. Defensible Absolute Uncertainty Bounds From Primary Sources

Reading each source directly and applying the GUM convention to unqualified figures, the defensible global 1σ net TOA uncertainty bounds are given in Table A9.

[88] does not state a single comprehensive global 1σ uncertainty of the net TOA. The calibration-only global 1σ uncertainty of 2 W m^{-2} is the paper's own stated figure and is a lower bound on the full global uncertainty, which must include radiance-to-flux conversion, time interpolation, and spectral response components not folded into that 2 W m^{-2}.

Table A8. Summary of the citation chain behind the IPCC AR6 "observed" Earth Energy Imbalance claim. Each row documents one quantitative claim, where it appears in the primary literature, what source it is attributed to, and what that source actually states.

Claim	Where it appears	Cited to	What the source actually says
"2% SW, 1% TOT at 95% CI"	Loeb 2009 [91] Section 4 and Appendix	[101]	Not in Priestley 2002 [101]. Numbers are Wielicki 1996 [130] goals × 2. [101] reports 0.33% SW / 0.20% LW achieved.
"0.85 ± 0.15 W m^{-2}"	Loeb 2009 [91], AR5 (2013) [59]	[49] model	Model target replaced CERES data. Admitted in Hansen 2011 [48].
"EEI = 0 ± 5.0 W m^{-2}"	L'Ecuyer 2015 [85]	Primary measurement	Retraction of Stephens 2012 [121]. Ignored by AR6.
"0.71 ± 0.10 W m^{-2} observed imbalance"	Loeb 2018 [88]	Johnson 2016 [66]	Argo-derived ocean heat uptake sum. Not a CERES satellite measurement.
"1% SW, 0.75% LW" (1σ)	Loeb 2018 [88], p. 897	No citation	LW loosened 50% from Wielicki goal. 3.75 times looser than Priestley 2002 achieved. Value picked to fit the required LW adjustment inside 1σ.
"1 W m^{-2} radiance-to-flux conversion"	Loeb 2018 [88] Section 4a	Su 2015 [123]	Correctly cited; Su 2015 is an ADM validation paper.
"1 W m^{-2} calibration (1σ)"	Loeb 2018 [88] Section 4a	No citation	Derived from 1% × 100 W m^{-2} SW stated uncited in Introduction.
"imbalance 0.7 (0.5, 0.9)" (W m^{-2})	AR6 (2021) [61]	No citation	This number and range appears nowhere in any of the quoted sources.

Table A9. Defensible absolute 1σ global net TOA uncertainty bounds from primary sources, with the basis for each value.

Source	1σ global net TOA uncertainty (W m^{-2})	Basis
Priestley 2002 [101] strictest (traceability only)	0.58	Early-mission FM-1 absolute traceability, SW + LW RSS
Priestley 2002 [101] combined RSS	1.85	Traceability + stability + unfiltering + FM-1/FM-2 agreement
Loeb 2018 [88] Introduction (p. 897)	2.00	Calibration only, stated as 1σ, lower bound on total
Priestley 2011 [102] FM-1/FM-2 structural floor	2.24	Direct same-platform residual, Edition 2, structural
L'Ecuyer 2015 [85] independent reanalysis	5.0	Combined satellite and OHCA
Stephens and L'Ecuyer 2015 [120] joint	5.6	Combined satellite and OHCA, *mea culpa* reanalysis

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synthesis, and iterative refinement (including critical independent reviews) of the manuscript through detailed analytical exchanges.

The AI systems are not listed as authors because they cannot take responsibility for the work, cannot approve the final version, and cannot be held accountable for its content, consistent with ICMJE, COPE, and WAME guidance. The AI systems were used as tools under full human direction. The authors reviewed, edited, revised, and approved every word of this manuscript. All arguments, analyses, and conclusions were independently evaluated and confirmed by the human authors, who accept full and sole responsibility for the accuracy, integrity, and originality of the published work.

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